# Digital IC Project : Repeater 설계 및 해석

## $R_{on}$ - $C_L$ 기반  CMOS Repeater PPA 분석

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# Outline

1. Project introduction

2. Summary

3. β-ratio

4. Rise / Fall time

5. Propagation delay

6. Duty cycle

7. Power consumption

8. Conclusion

# Project Introduction

# Project Objective

**Same Load**와 **Multi Load** 조건의 Repeater 회로를 설계하고,
**PVT**(Process / Voltage / Temperature) 변화에 따른 특성을 비교한다.
이를 $R_{on}$, $C_L$ 관점서 해석하여 **PPA**(Performance/Power/Area)를 평가한다.

**공정 (Process Technology)**
45 nm / 90 nm / 180 nm

**트랜지스터 크기 (Multiplier)**
1X > M=1 / 2X > M=2 / 4X > M=4

**공급 전압 (Voltage)**
45 nm: 1.1V / 90 nm: 1.2V / 180 nm: 2.0V

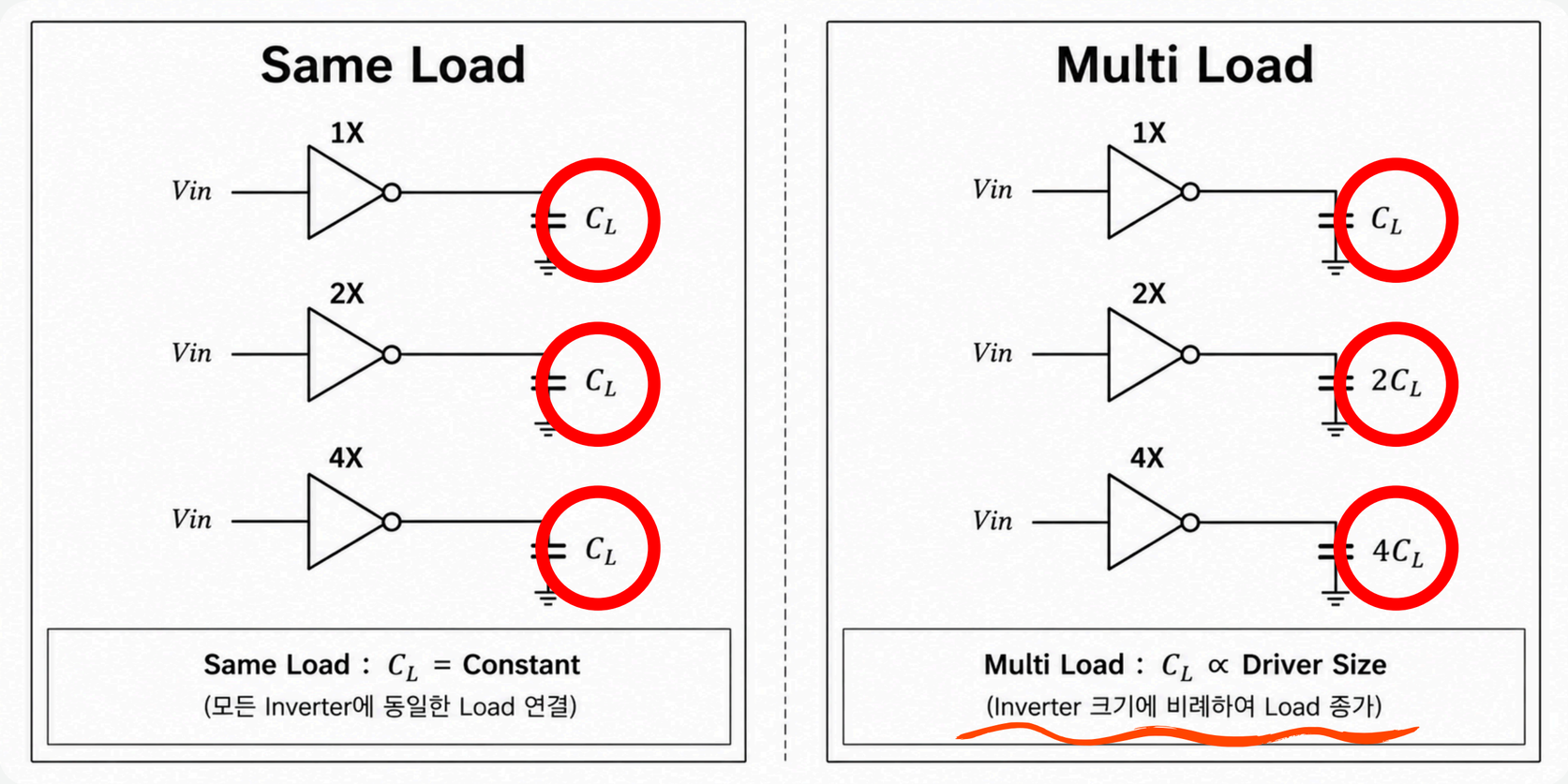
**Corner Condition**
SS 85°C (Worst) / NN 27°C (Typical) / FF -25°C (Best)

**부하 조건 (Load Condition)**
Same Load / Multi Load

p. 5 참조

# Repeater, Same/Multi Load

**Driver**가 부하를 안정적으로 구동할 수 있도록 중간에 **Inverter**를 삽입하여
**신호**의 전달 지연을 **개선**하는 회로



# $R_{on}$ & $C_L$ (1/3)

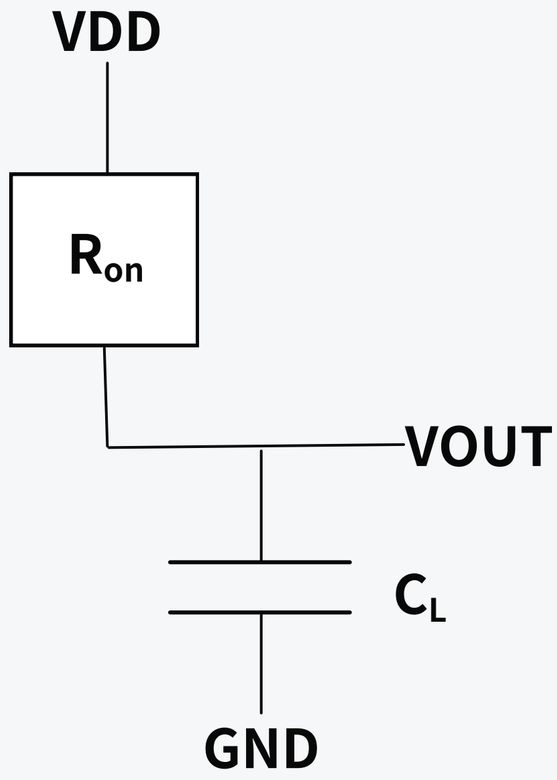
- **RC model of a repeater (inverter)**

• Repeater의 동작을 단순화하면,
ON 상태의 transistor는 등가 저항 $R_{on}$으로,
출력에 연결된 모든 부하는 등가 capacitance $C_L$로 표현할 수 있다.

• 이 단순화된 1차 RC 모델에서
propagation delay는 $R_{on}$과 $C_L$의 곱에 비례한다.

**Propagation Delay $\propto$ $R_{on} \cdot C_L$**

- **$R_{ON}$, $C_L$ 이란?**




**$R_{on}$**

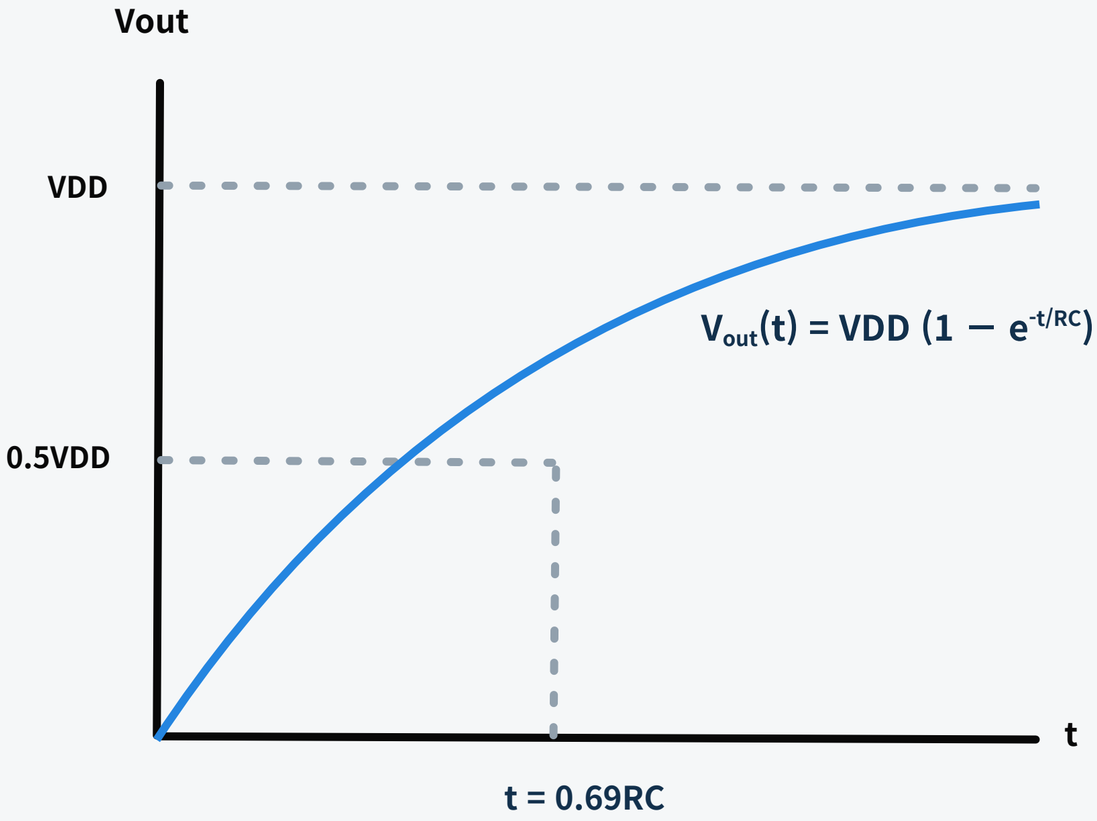
• Transistor의 ON 저항
• PMOS, NMOS 각각 존재 (Rp, Rn)
• Transistor Width 클수록
구동 전류 증가→ $R_{on}$↓ ($R_{on} = V_{DS}/I_D$)

**$C_L$ (Load Capacitance)**

• Driver 내부의 기생 C
• 배선 및 연결에 의한 C
• 다음 Load에서 보이는 C

# $R_{on}$ & $C_L$ (2/3)

- **RC charging / discharging**
  - 저항을 통해 capacitor를 충전(또는 방전)할 때 출력 전압은 지수적으로 변화한다.




- **Time Constant**

$$\tau = R_{on} \cdot C_L$$

- 출력 전압이 얼마나 빠르게 변하는지를 결정하는 시간 척도
- 예시: $R_{on}$ = 1 kΩ, $C_L$ = 10 fF → τ = 10 ps

- **RC delay - 왜 $0.69 \cdot R_{on} \cdot C_L$인가?**

$$t = \tau \ln 2 \approx 0.69 \cdot R_{on} \cdot C_L$$

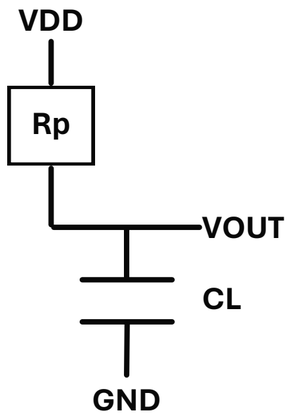
- $V_{out}$ = 0.5VDD일 때의 시간이 propagation delay로 정의
- 0.69 = ln2 ≈ 0.693

# $R_{on}$ & $C_L$ (3/3)

- **Two transition paths in CMOS inverter**

### Output Low → High ($t_{pLH}$)

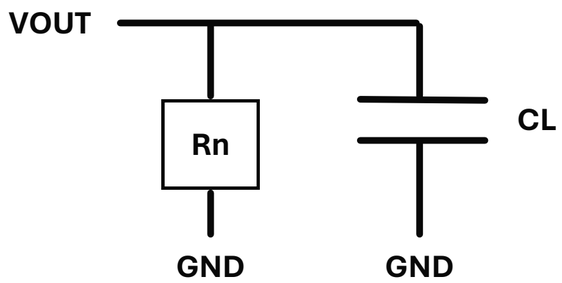
• PMOS가 ON 되어 CL을 VDD로 충전




$$t_{pLH} \approx 0.69 \cdot R_p \cdot C_L$$

### Output High → Low ($t_{pHL}$)

• NMOS가 ON 되어 CL을 GND로 방전




$$t_{pHL} \approx 0.69 \cdot R_n \cdot C_L$$

- **Total Propagation Delay**

  상승 및 하강 지연 시간의 평균으로 정의

$$t_{pd} = (t_{pHL} + t_{pLH}) / 2$$

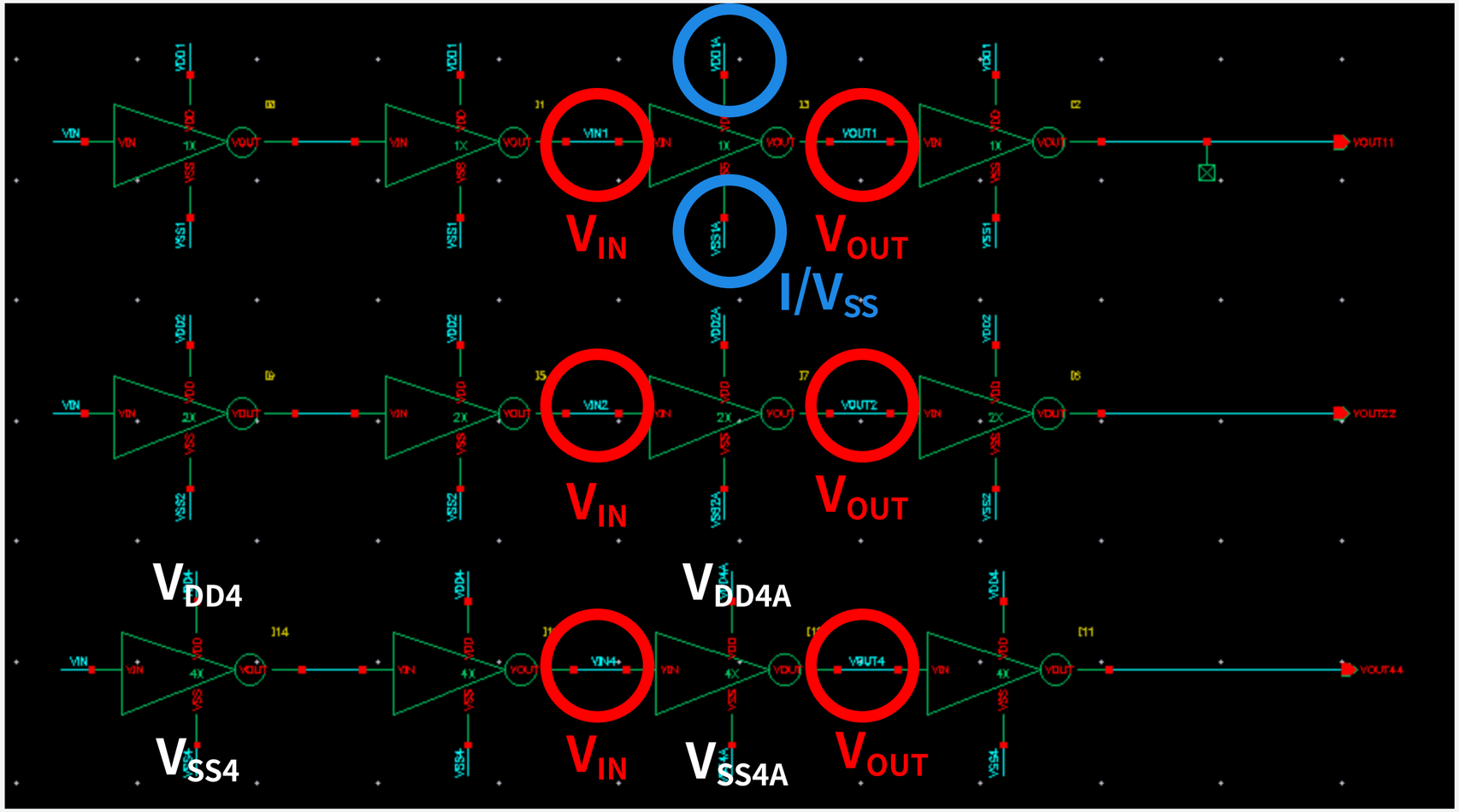
$$\approx 0.69 \cdot (R_n + R_p)/2 \cdot C_L$$

- **Design Implication**

• $R_{on}$ ↑ 또는 $C_L$ ↑ → $t_{pd}$ ↑
• Transistor를 크게 설계하면 $R_{on}$ ↓ → $t_{pd}$ ↓
• 하지만 기생 capacitance도 증가할 수 있어 trade-off 고려 필요

# Measurement Point on Schematic

- 3번째 Inverter를 Driver로 설정 $I/V_{DD}$

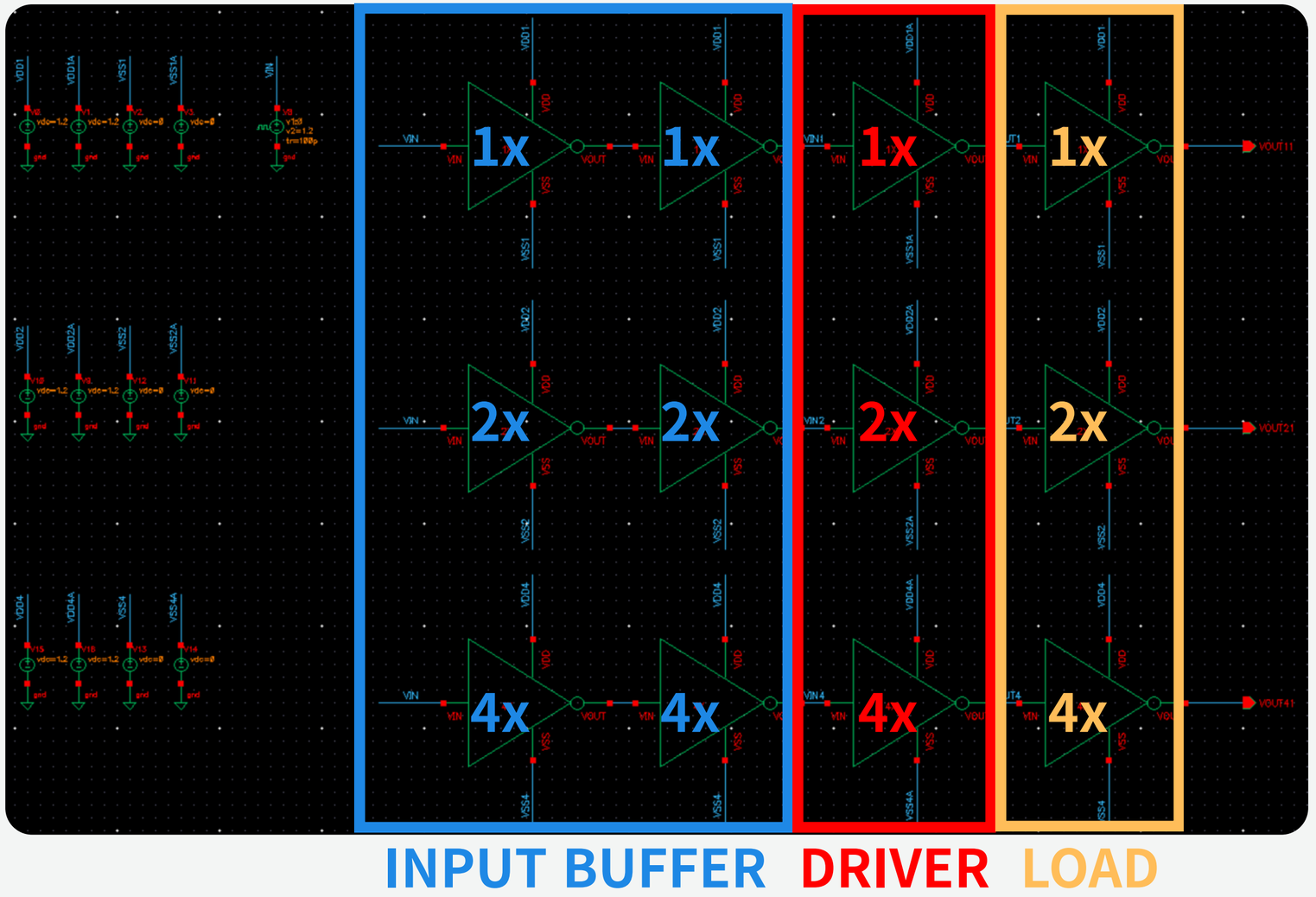



## Power Supply 분리

- 각 Inverter의 $V_{DD}$/ $V_{SS}$ 전원 경로 분리
- Stage 간 전원 경로의 상호 영향 최소화

# Schematic

- Schematic of Repeater with Same Load



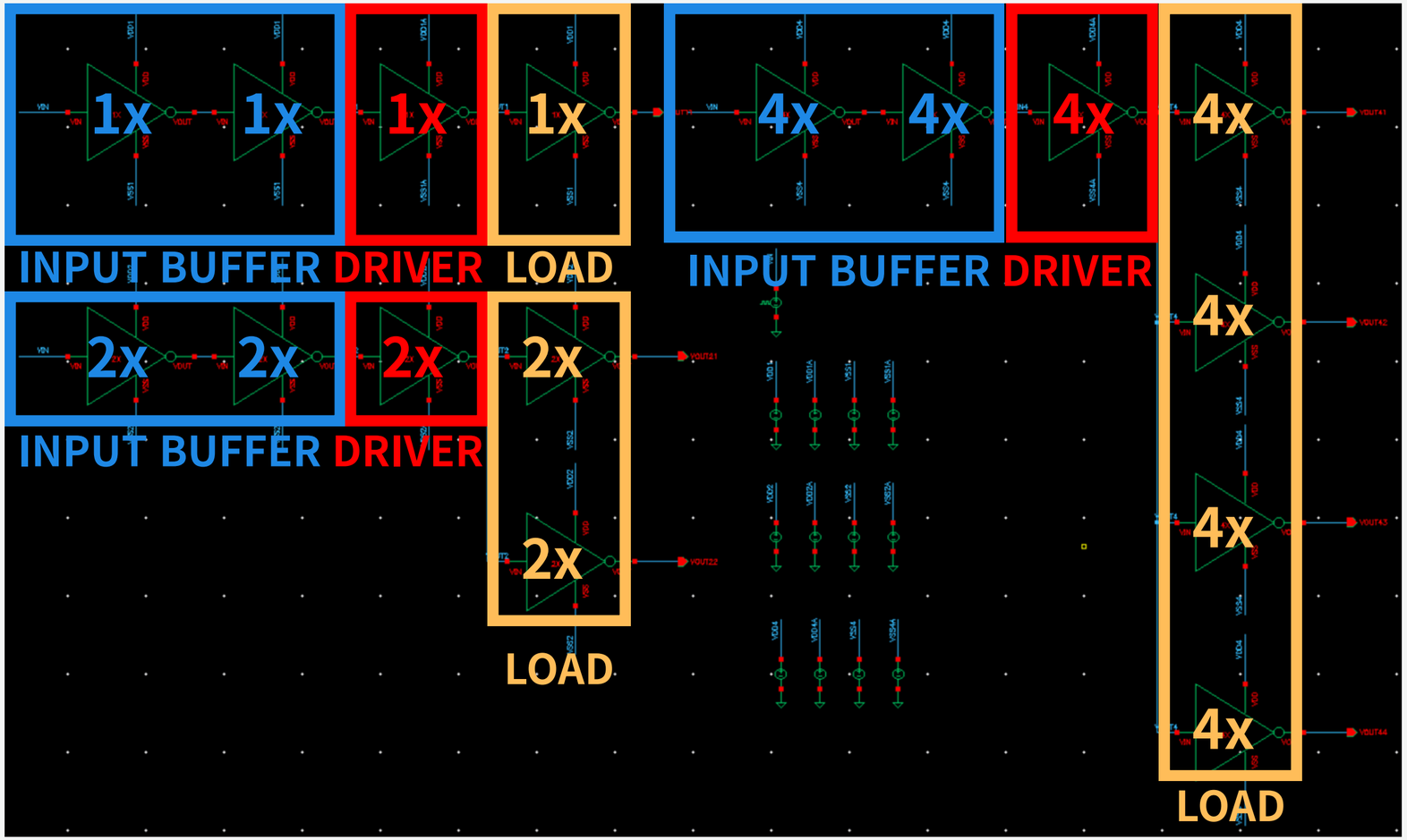

1X -> M=1

2X -> M=2

4X -> M=4

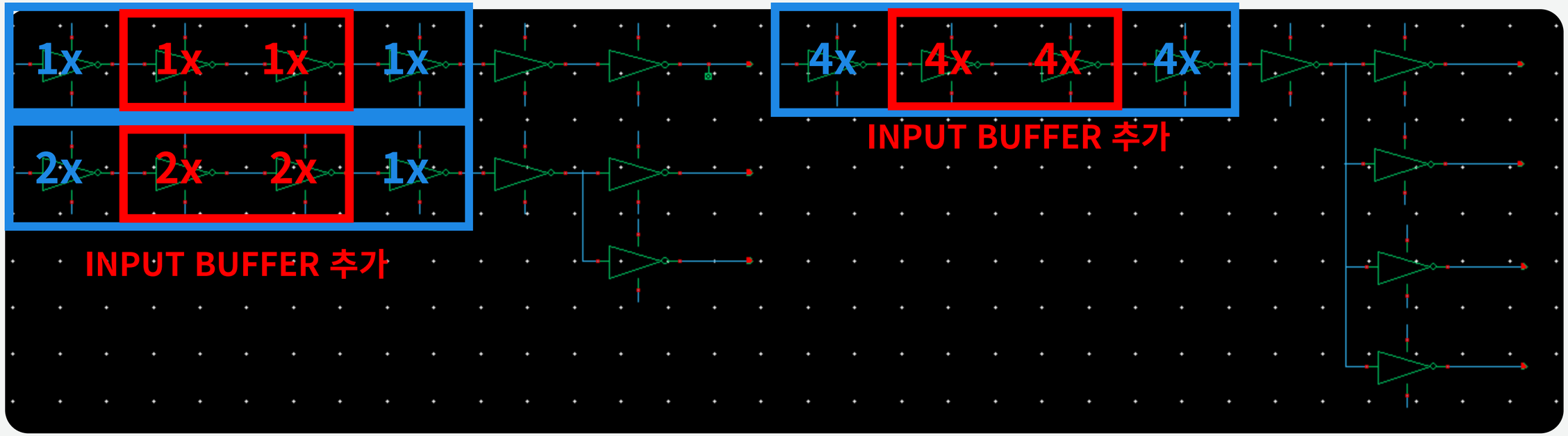
# Schematic

- Schematic of Repeater with Multi Load



# Schematic

- Schematic of Repeater with gpdk090 + 2 INVERTER




- **Input Buffer 추가 목적**

  1) Driver에 인가되는 입력 slew의 영향 감소
  2) 일정한 입력 조건에서 Driver 자체의 Delay 특성 확인

# Simulation Conditions

| Process [nm] | Voltage [V] | VLow [V] | NMOS | | PMOS | | Corner / Temp [℃] | | | Period [ns] | trise [ps] | tfall [ps] | Pulse width [ps] |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | L [nm] | W [nm] | L [nm] | W [nm] | SS | NN | FF | | | | |
| 45 | 1.1 | 0 | 45 | 120 | 45 | 140 | 85 | 27 | -25 | 1 | 100 | 100 | 400 |
| 90 | 1.2 | 0 | 100 | 120 | 100 | 350 | | | | | | | |
| 180 | 2 | 0 | 180 | 400 | 180 | 1800 | | | | | | | |

**NMOS width → 각 공정 최솟값 / PMOS width → β-ratio 시뮬레이션**
**Worst case: SS, 85℃ / Typical case: NN, 27 ℃ / Best case: FF, -25 ℃**

**※ 변화를 명확하게 관찰하기 위해 boundary condition으로 Process corner와 온도를 함께 설정**

# Summary

# Summary

| PPA & Parameters | Process Node | Multiplier | | Corner |
|---|---|---|---|---|
| | 180nm → 90nm → 45nm | 1x → 2x → 4x | | SS(85℃) → NN(27℃) → FF(-25℃) |
| | | Same load | Multi load | |
| Power<br>(IDD, ISS) | 감소 | 증가 | | 감소 |
| Performance<br>(Rise&Fall, Propagation Delay) | 증가 | 변화 X | 감소 | 증가 |
| Area | 감소 | 증가 | | 변화 X |
| PPA analysis | Power 감소<br>Performance 향상<br>Area 감소 | Power 증가<br>Performance 동일<br>Area 증가 | Power 증가<br>Performance 감소<br>Area 증가 | Power 감소<br>Performance 향상<br>Area 동일 |

# β-Ratio

# β-Ratio

- NMOS와 PMOS의 구동력(strength) 비율

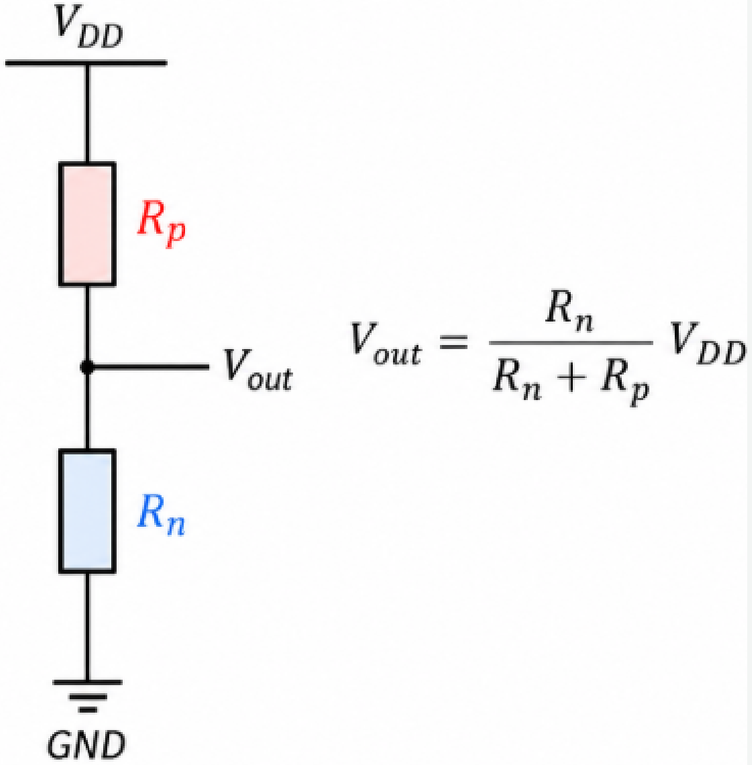
  NMOS와 PMOS의 구동력(strength) 차이를 width로 보정하는 것

$$\beta_n = \mu_n \cdot C_{ox} \cdot (W_n / L_n)$$
$$\beta_p = \mu_p \cdot C_{ox} \cdot (W_p / L_p)$$
$$\beta_p / \beta_n = [\mu_p \cdot C_{ox} \cdot (W_p/L_p)] / [\mu_n \cdot C_{ox} \cdot (W_n/L_n)]$$
$$\beta_p / \beta_n = (\mu_p/\mu_n) \cdot (W_p/W_n) \cdot (L_n/L_p)$$
$$\beta_p / \beta_n \approx (\mu_p/\mu_n) \cdot (W_p/W_n)$$

$$\beta_p / \beta_n \approx 1$$

**PMOS Pull-up strength ≈ NMOS Pull-down strength**

VTC(Voltage Transfer Characteristic) 그래프를 통해
$W_n$ 은 특정 값으로 고정하고 $W_p$ 를 Sizing 변수로 시뮬레이션하여
$V_m \approx VDD/2$ 인 $W_p$의 값 도출



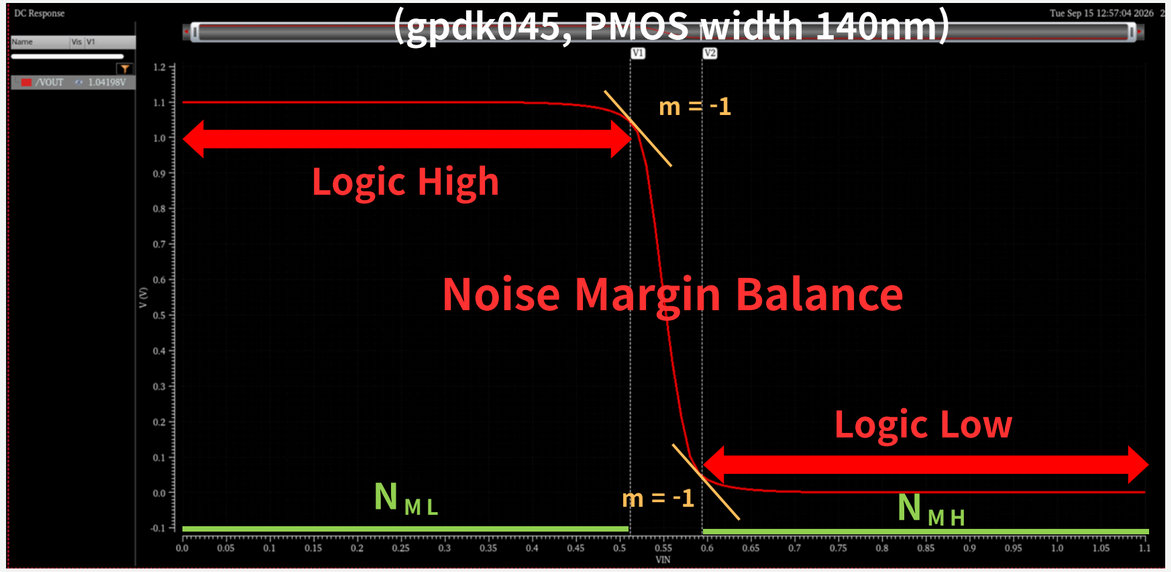

$$\frac{\beta_p}{\beta_n} = \frac{\mu_p C_{ox} \frac{W_p}{L_p}}{\mu_n C_{ox} \frac{W_n}{L_n}} = \frac{\mu_p}{\mu_n} \cdot \frac{W_p}{W_n} \cdot \frac{L_n}{L_p}$$

$L_n = L_p$ 라고 가정하면, $\frac{\beta_p}{\beta_n} \approx \frac{\mu_p}{\mu_n} \cdot \frac{W_p}{W_n}$

# β-Ratio - Noise Margin

- 회로가 0,1을 정상적으로 인식할 수 있는 범위

  VTC에서 기울기가 -1인 지점까지 회로가 0,1을 정상적으로 인식할 수 있는 범위이며, 이를 기준으로 $V_{IL}$과 $V_{IH}$를 정의




**$\beta_p/\beta_n \approx 1$ 일수록 VTC의 switching point가 VDD/2에 가까워지며, $N_{ML}$과 $N_{MH}$도 균형을 이룰 수 있음**

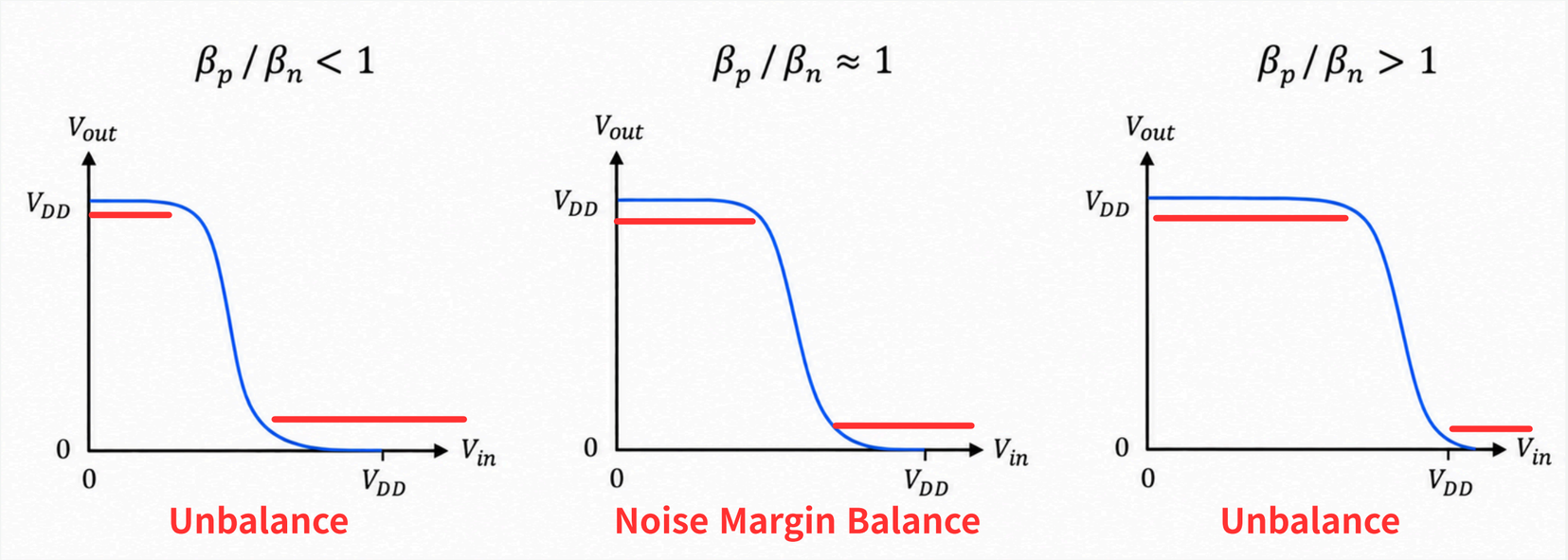
$$\frac{\beta_p}{\beta_n} \approx 1$$

$$V_M \approx \frac{V_{DD}}{2}$$

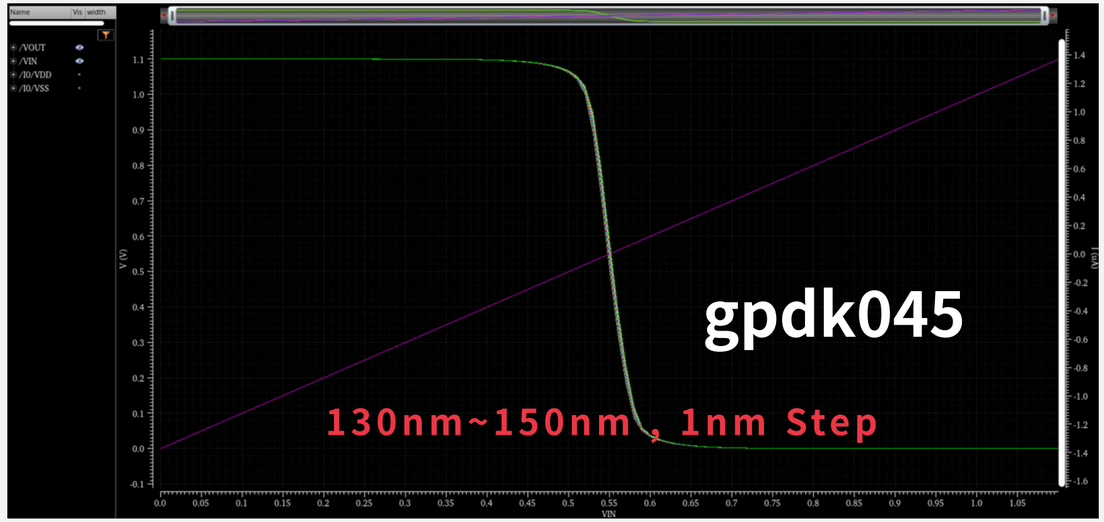
$$N_{ML} \approx N_{MH}$$

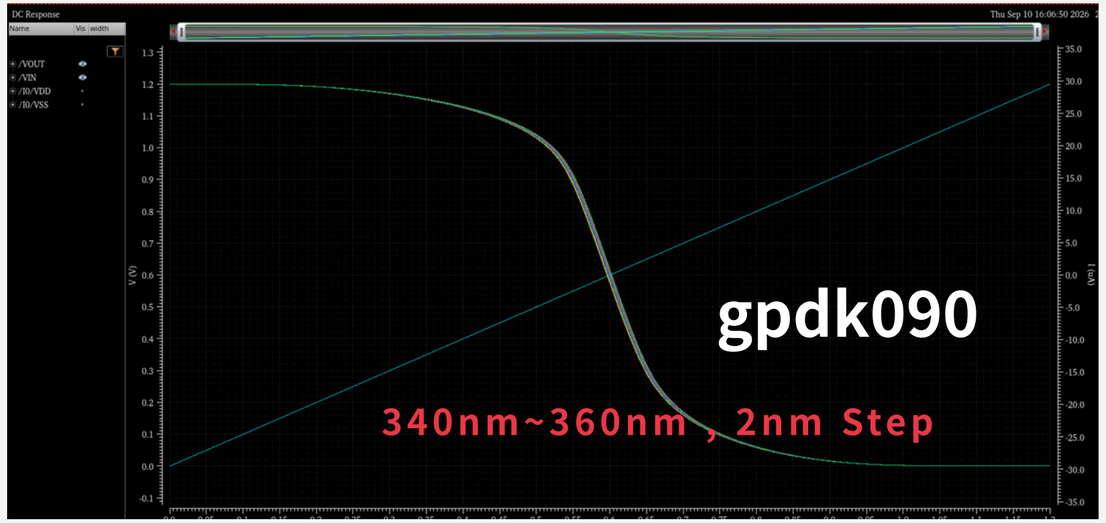
$$\frac{\beta_p}{\beta_n} \uparrow \Rightarrow V_M \uparrow \Rightarrow N_{ML} \uparrow,\ N_{MH} \downarrow$$

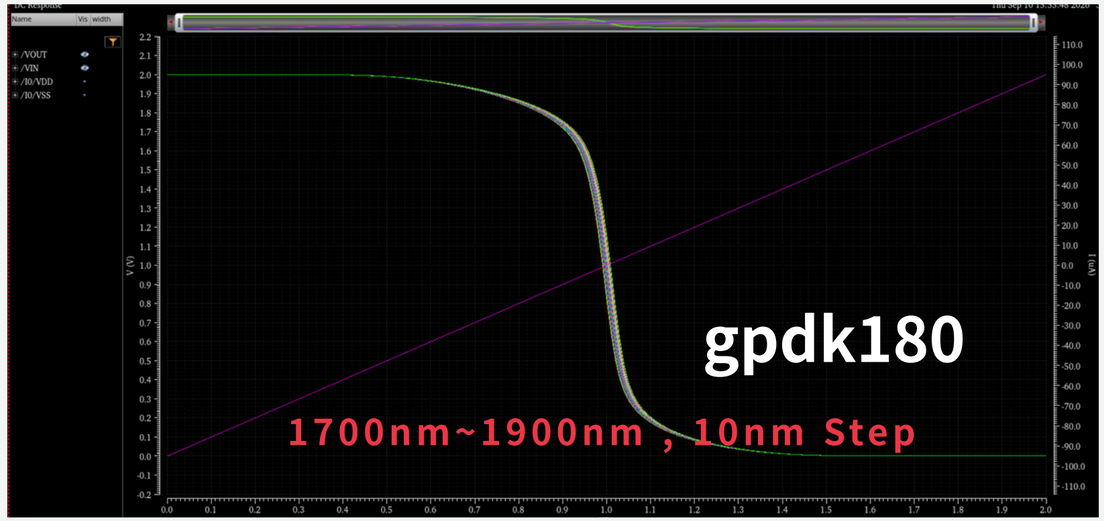
$$\frac{\beta_p}{\beta_n} \downarrow \Rightarrow V_M \downarrow \Rightarrow N_{ML} \downarrow,\ N_{MH} \uparrow$$



# β-Ratio - VTC of gpdk045, 090, 180(PMOS width sweep)




[gpdk045] - Range: 130nm~150nm , Sweep Step: 1nm

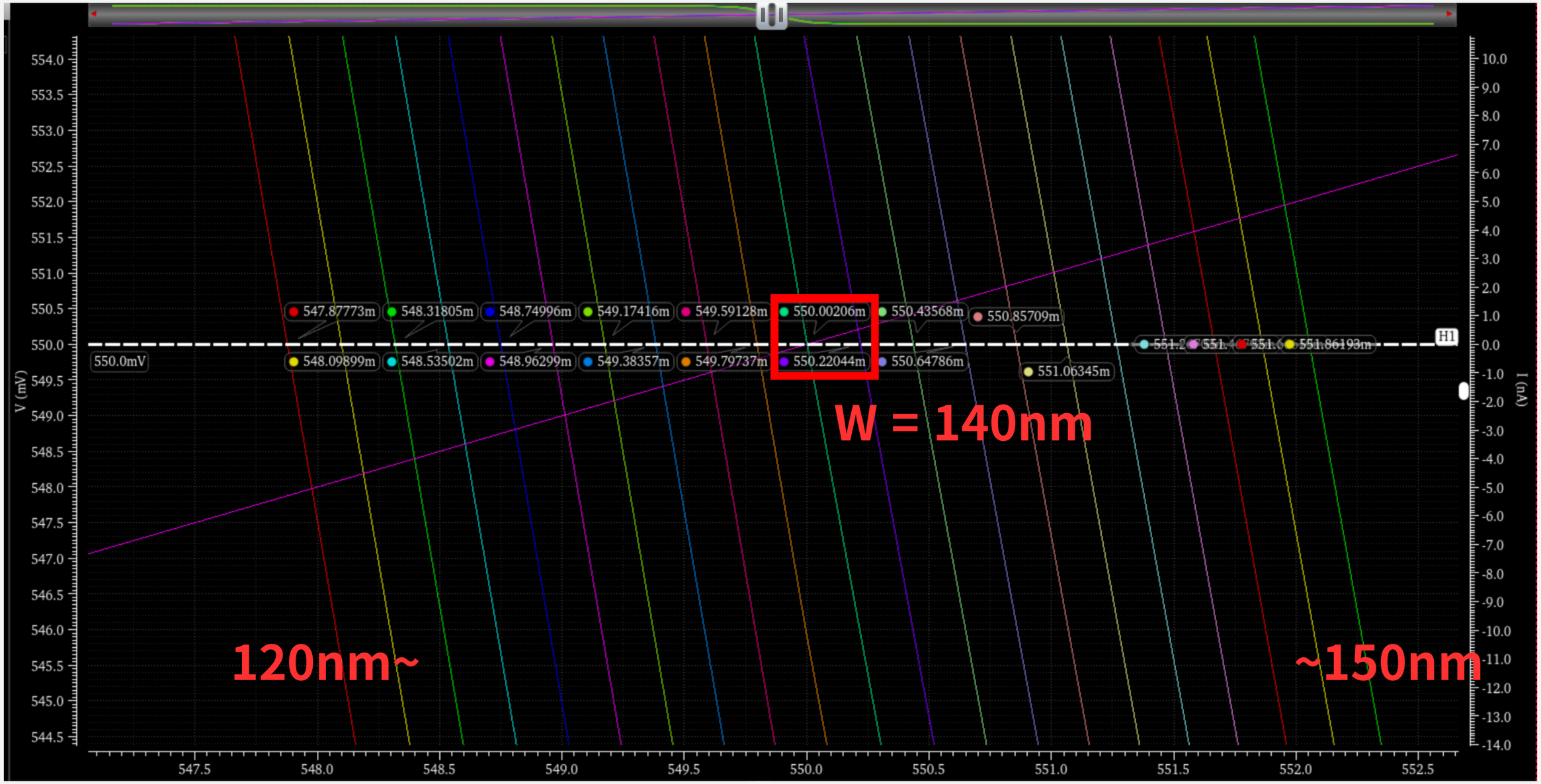
[gpdk090] - Range: 340nm~360nm , Sweep Step: 2nm

[gpdk180] - Range: 1700nm~1900nm , Sweep Step: 10nm

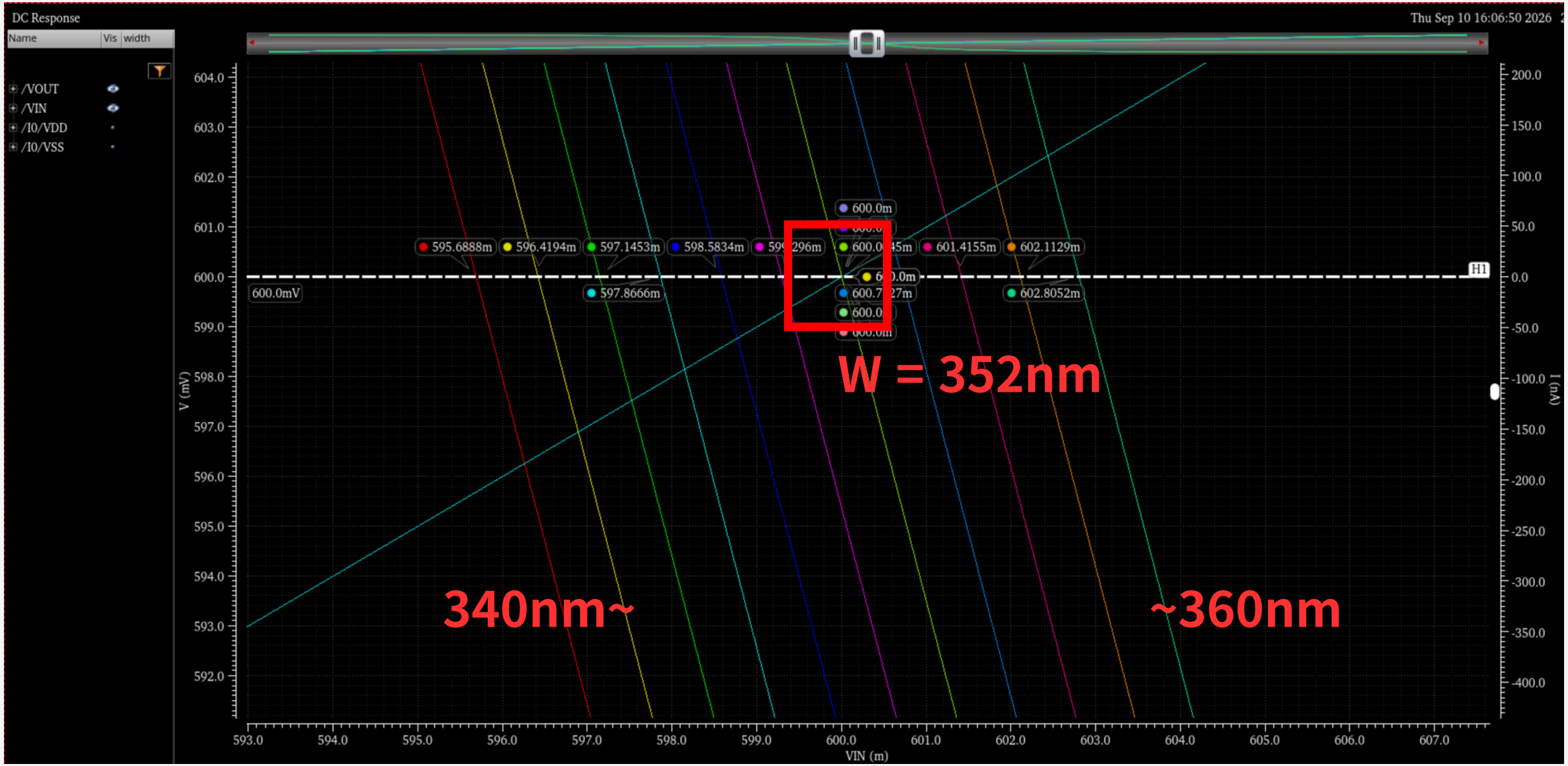
**$W_n$ Fixed, $W_p$ Simulation → $V_m \approx VDD/2$**

*$W_n$ = 45nm(045), 100nm(090), 180nm(180)

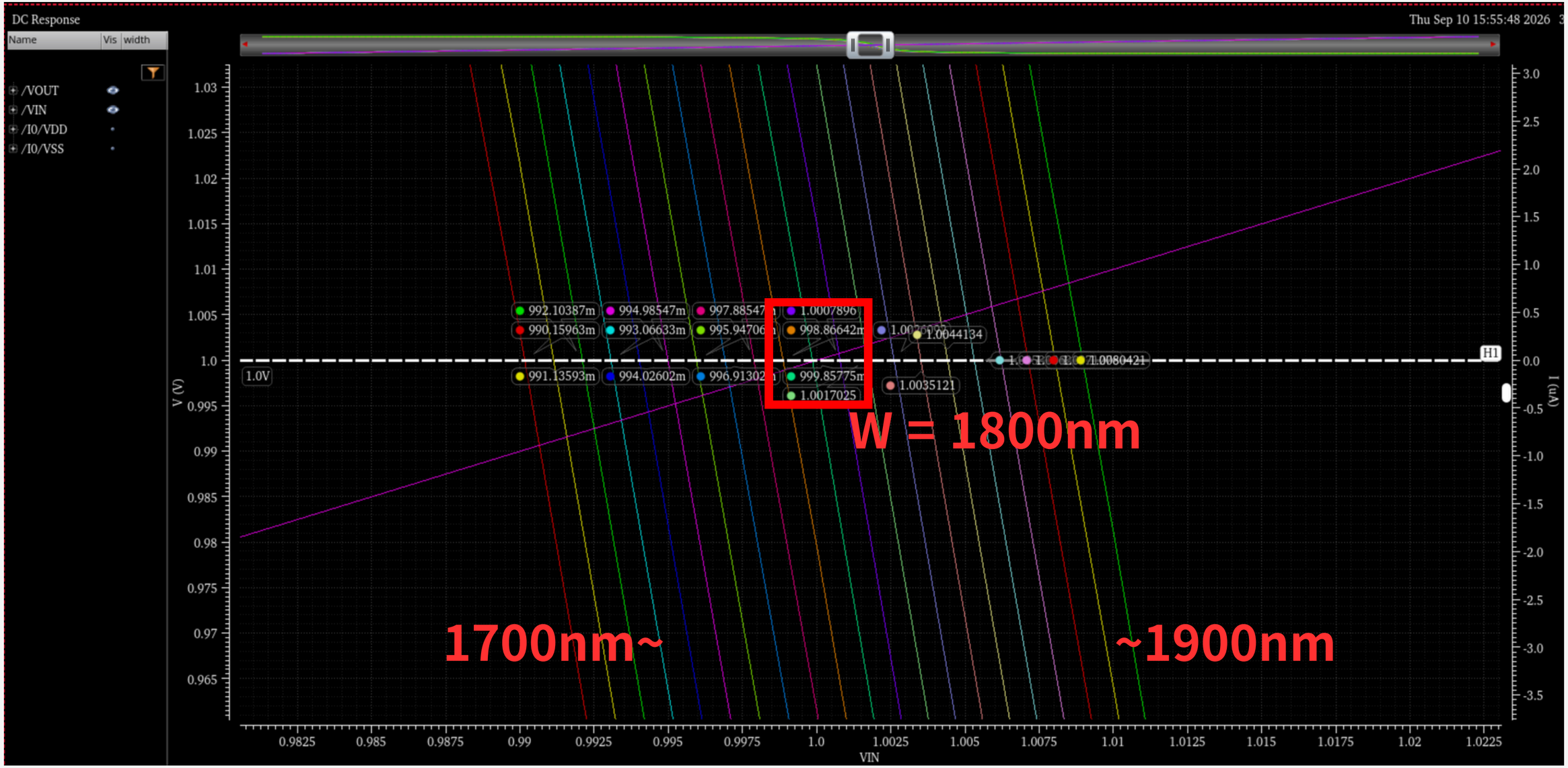
# β-Ratio - VTC of gpdk045(PMOS width sweep)



# β-Ratio - VTC of gpdk090(PMOS width sweep)



# β-Ratio - VTC of gpdk180(PMOS width sweep)



# β-Ratio - Result

- 공정 별 NMOS/PMOS strength 차이 → ($\beta_p$ / $\beta_n$ ≈ 1)을 위한 $W_p/W_n$ 비율 공정마다 상이

  공정별 NMOS와 PMOS의 구동 특성이 다르므로,
  Pull-down과 Pull-up strength를 균형화하기 위한 $W_p/W_n$ 또한 공정마다 달라짐
  → 해당 시뮬레이션에서는 공정 Size가 증가할수록 β balance를 위한 PMOS width 증가

| Process Node [nm] | Supply Voltage ($V_{DD}$) [V] | NMOS (n-channel) | | PMOS (p-channel) | | $W_p / W_n$ [-] |
|---|---|---|---|---|---|---|
| | | $L_n$ [nm] | $W_n$ [nm] | $L_p$ [nm] | $W_p$ [nm] | |
| **45** | 1.1 | 45 | 120 | 45 | 140 | **1.17** |
| **90** | 1.2 | 100 | 120 | 100 | 350 | **2.92** |
| **180** | 2.0 | 180 | 400 | 180 | 1800 | **4.50** |

※ 90 nm 공정의 실제 시뮬레이션 결과 $W_p$ = 352 nm 적용 시, $W_p/W_n$ = 2.93

# Rise & Fall

# Rise & Fall

- 출력 신호가 Low ↔ High 상태로 전환되는 데 걸리는 시간

CMOS inverter의 출력 노드는 Load Capacitance $C_L$의 충·방전에 의해 순간적으로 변하지 않으며,
PMOS와 NMOS의 구동 능력 및 $C_L$의 크기에 따라 Rise/Fall Time이 결정된다.

## $t_r$ (Rise Time)

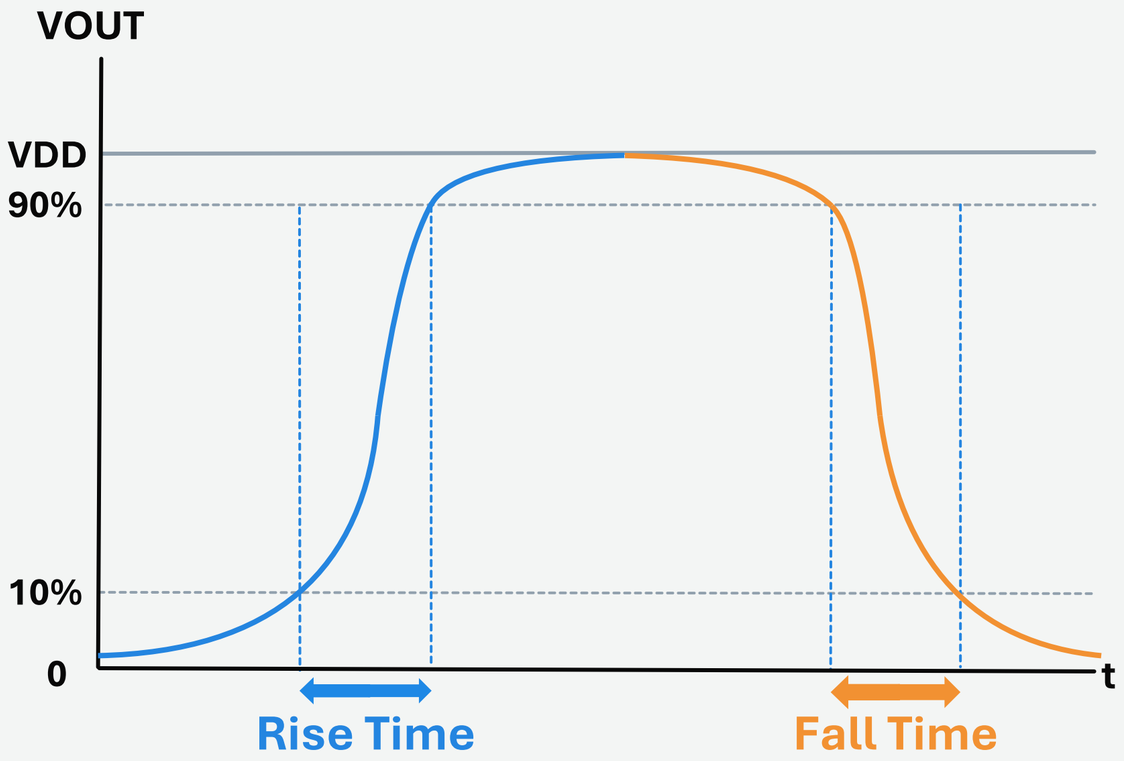
- **Output : Low → High**
- **출력 전압이 10% 지점에서 90% 지점까지 도달하는 데 걸리는 시간**

$$t_r \approx \frac{C_L}{\beta_p V_{DD}} = \frac{C_L}{\mu_p C_{ox} V_{DD}} \left(\frac{L_p}{W_p}\right) \approx 2.2 R_{on,p} C_L$$

## $t_f$ (Fall Time)

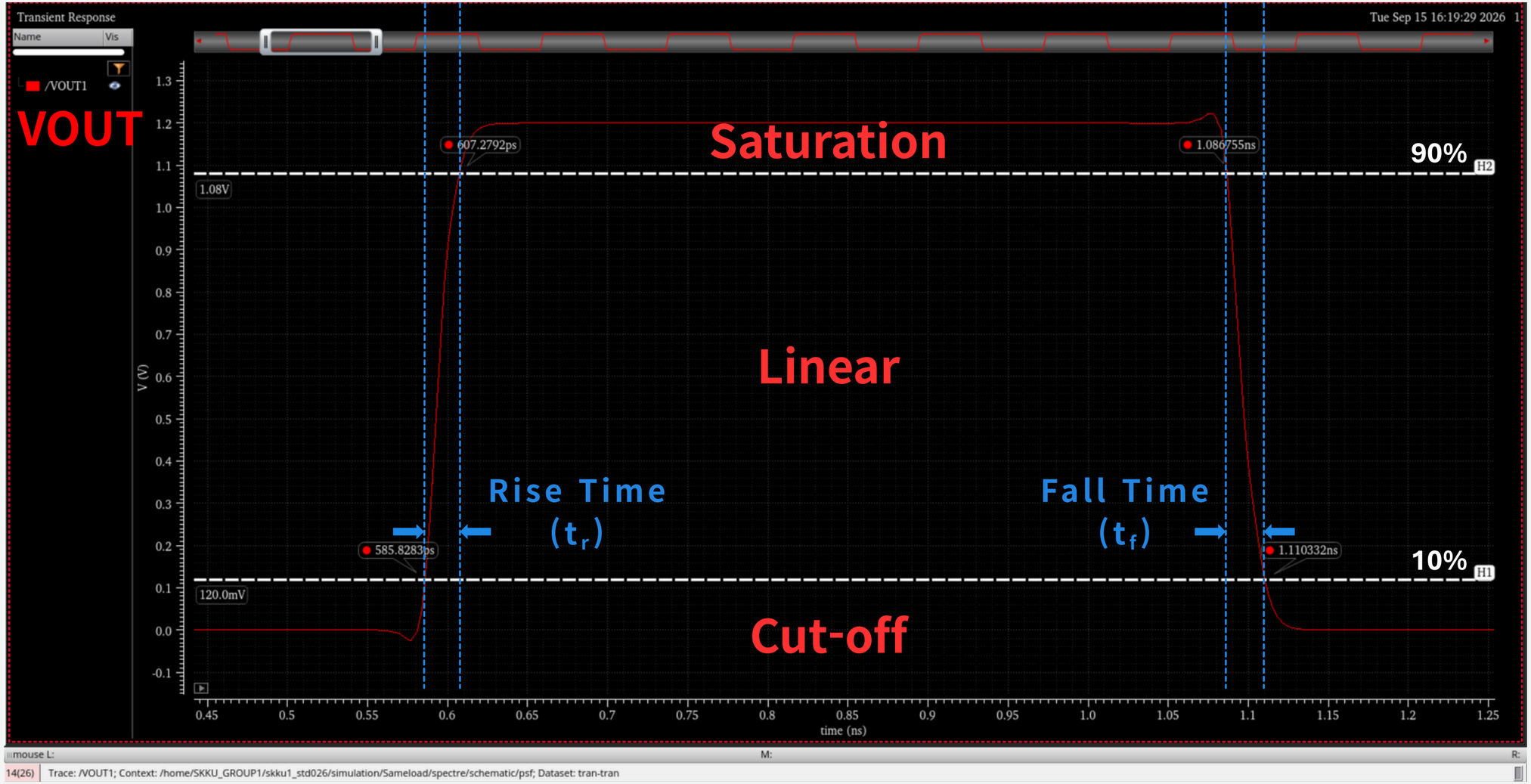
- **Output : High → Low**
- **출력 전압이 90% 지점에서 10% 지점까지 도달하는 데 걸리는 시간**

$$t_f \approx \frac{C_L}{\beta_n (V_{DD} - V_{TN})} = \frac{C_L}{\mu_n C_{ox} \left(W_n/L_n\right)(V_{DD} - V_{TN})} \approx 2.2 R_{on,n} C_L$$



# Rise & Fall

[ Transient simulation result – VOUT ]



# Same Load

| Process Tech [nm] | Multiplier factor | Process Corner | | | | | |
|---|---|---|---|---|---|---|---|
| | | SS(85°C) | | NN(27°C) | | FF(-25°C) | |
| | | $t_r$ - Rise [ps] | $t_f$ - Fall [ps] | $t_r$ - Rise [ps] | $t_f$ - Fall [ps] | $t_r$ - Rise [ps] | $t_f$ - Fall [ps] |
| gpdk 180 | 1x | 64.78 | 73.60 | 51.07 | 54.33 | 40.14 | 42.30 |
| | 2x | 64.78 | 73.60 | 51.07 | 54.33 | 40.14 | 42.30 |
| | 4x | 64.78 | 73.60 | 51.07 | 54.33 | 40.14 | 42.30 |
| gpdk 090 | 1x | 35.59 | 43.03 | 21.45 | 23.65 | 14.5 | 14.91 |
| | 2x | 35.59 | 43.03 | 21.45 | 23.65 | 14.5 | 14.91 |
| | 4x | 35.59 | 43.03 | 21.45 | 23.65 | 14.5 | 14.91 |
| gpdk 045 | 1x | 12.92 | 14.04 | 8.96 | 11.00 | 6.66 | 8.57 |
| | 2x | 12.92 | 14.04 | 8.96 | 11.00 | 6.66 | 8.57 |
| | 4x | 12.92 | 14.04 | 8.96 | 11.00 | 6.66 | 8.57 |
| gpdk090 +2 inv. | 1x | 35.46 | 42.81 | 20.95 | 23.26 | 13.90 | 14.36 |
| | 2x | 35.46 | 42.81 | 20.95 | 23.26 | 13.90 | 14.36 |
| | 4x | 35.46 | 42.81 | 20.95 | 23.26 | 13.90 | 14.36 |

① Process :
공정 미세화 → W/L, $V_{DD}$감소
$R_{ON}$ 증가 << $C_L$감소가 우세
결과적으로 $t_r$, $t_f$ 감소 → performance 향상

② Multiplier :
$R_{ON}$ → 1/N배 감소
$C_L$ → N배 증가
$t_r$, $t_f$ 일정 → performance 영향 없음

③ Corner :
SS > NN > FF 갈수록
$V_{th}$ 감소 → performance 향상

# Multi Load

| Process Tech [nm] | Multiplier factor | Process Corner | | | | | |
|---|---|---|---|---|---|---|---|
| | | SS(85°C) | | NN(27°C) | | FF(-25°C) | |
| | | $t_r$ - Rise [ps] | $t_f$ - Fall [ps] | $t_r$ - Rise [ps] | $t_f$ - Fall [ps] | $t_r$ - Rise [ps] | $t_f$ - Fall [ps] |
| gpdk 180 | 1x | 64.72 | 73.57 | 50.70 | 54.30 | 40.06 | 42.13 |
| | 2x | 92.93 | 111.04 | 73.50 | 82.70 | 57.71 | 64.48 |
| | 4x | 141.82 | 183.14 | 111.39 | 137.88 | 87.57 | 108.19 |
| gpdk 090 | 1x | 35.64 | 43.03 | 21.44 | 23.64 | 14.53 | 14.92 |
| | 2x | 50.89 | 67.11 | 30.54 | 36.82 | 20.93 | 23.16 |
| | 4x | 80.42 | 115.6 | 49.33 | 63.24 | 34.44 | 39.91 |
| gpdk 045 | 1x | 12.91 | 14.10 | 8.98 | 11.00 | 6.65 | 8.57 |
| | 2x | 19.84 | 22.23 | 13.96 | 16.83 | 10.24 | 12.80 |
| | 4x | 33.21 | 36.57 | 22.78 | 27.33 | 16.39 | 20.98 |
| gpdk090 +2 inv. | 1x | 35.46 | 42.81 | 20.95 | 23.26 | 13.90 | 14.36 |
| | 2x | 50.90 | 67.03 | 30.25 | 36.48 | 20.58 | 22.79 |
| | 4x | 80.39 | 115.01 | 49.05 | 63.11 | 34.23 | 39.55 |

① Process :
공정 미세화 → W/L, $V_{DD}$감소
$R_{ON}$ 증가 << $C_L$감소가 우세
결과적으로 $t_r$, $t_f$ 감소 → performance 향상

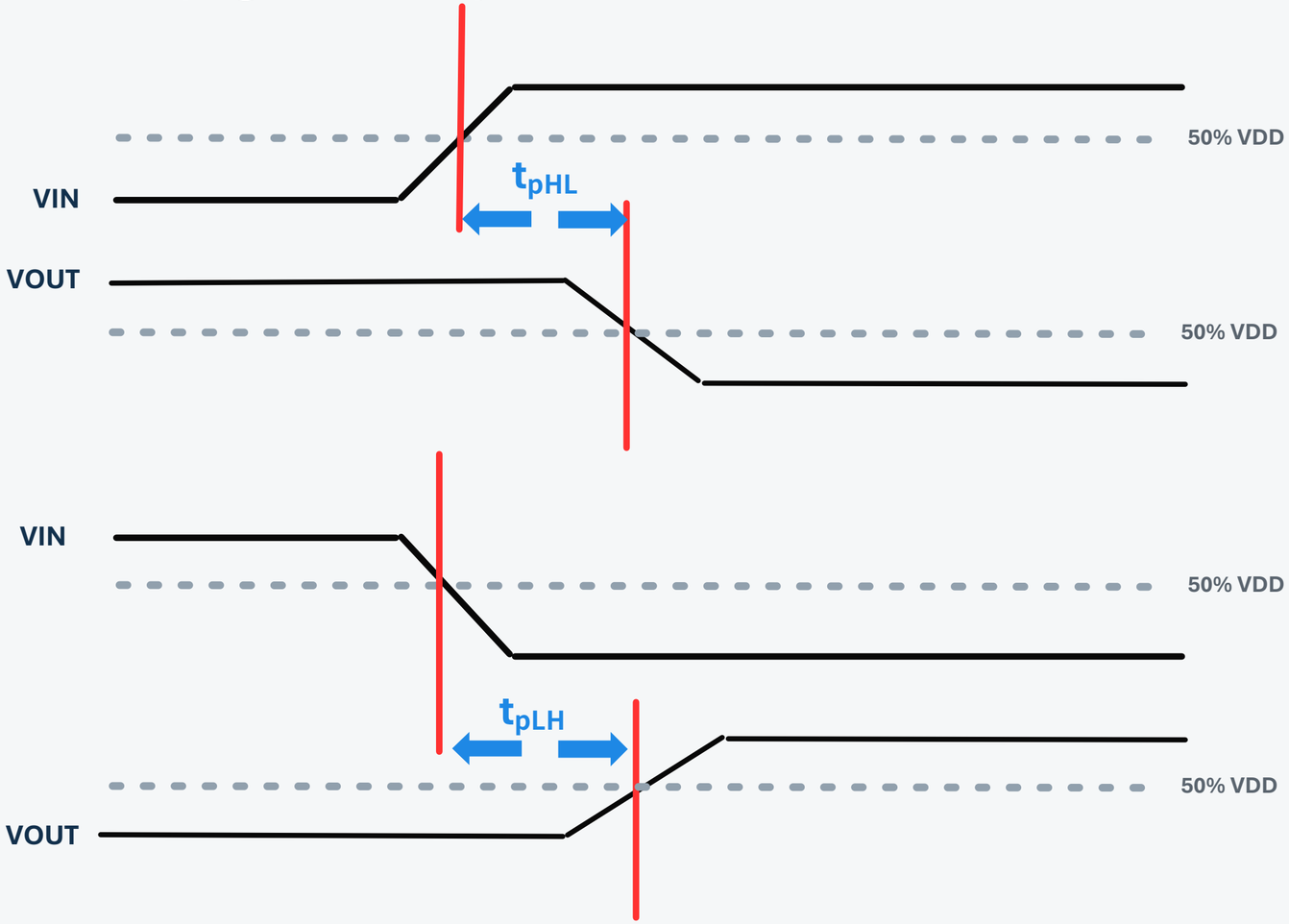
② Multiplier :
$R_{ON}$ → 1/N배 감소
$C_L$ → $(N+N^2)$배 증가
$t_r$, $t_f$ 증가 → performance 감소

③ Corner :
SS > NN > FF 갈수록 $V_{th}$ 감소
→ peformance 향상

# Propagation Delay

# Propagation Delay

- 입력 신호의 변화가 출력에 전달되기까지 걸리는 시간
- Propagation Delay는 입력과 출력의 50% VDD 지점 사이의 시간 차이로 측정




## 왜 50% VDD인가?

- 입력/출력 모두 동일한 기준으로 비교 가능
- Rise/Fall의 끝점을 사용하는 것보다 switching timing을 일관되게 정의할 수 있음
- 예: VDD = 1.2 V → 50% = 0.6 V
- tpHL과 tpLH를 각각 측정한 뒤 평균하여 tpd 계산

# Propagation Delay

CMOS inverter에서는 출력 Load Capacitance의 충·방전 과정 때문에
입력과 출력 사이에 시간 지연이 발생한다.

**$t_{pHL}$**

Input : Low → High
Output : High → Low

NMOS가 $C_L$을 방전시키는 과정의 지연

**$t_{pLH}$**

Input : High → Low
Output : Low → High

PMOS가 $C_L$을 충전시키는 과정의 지연

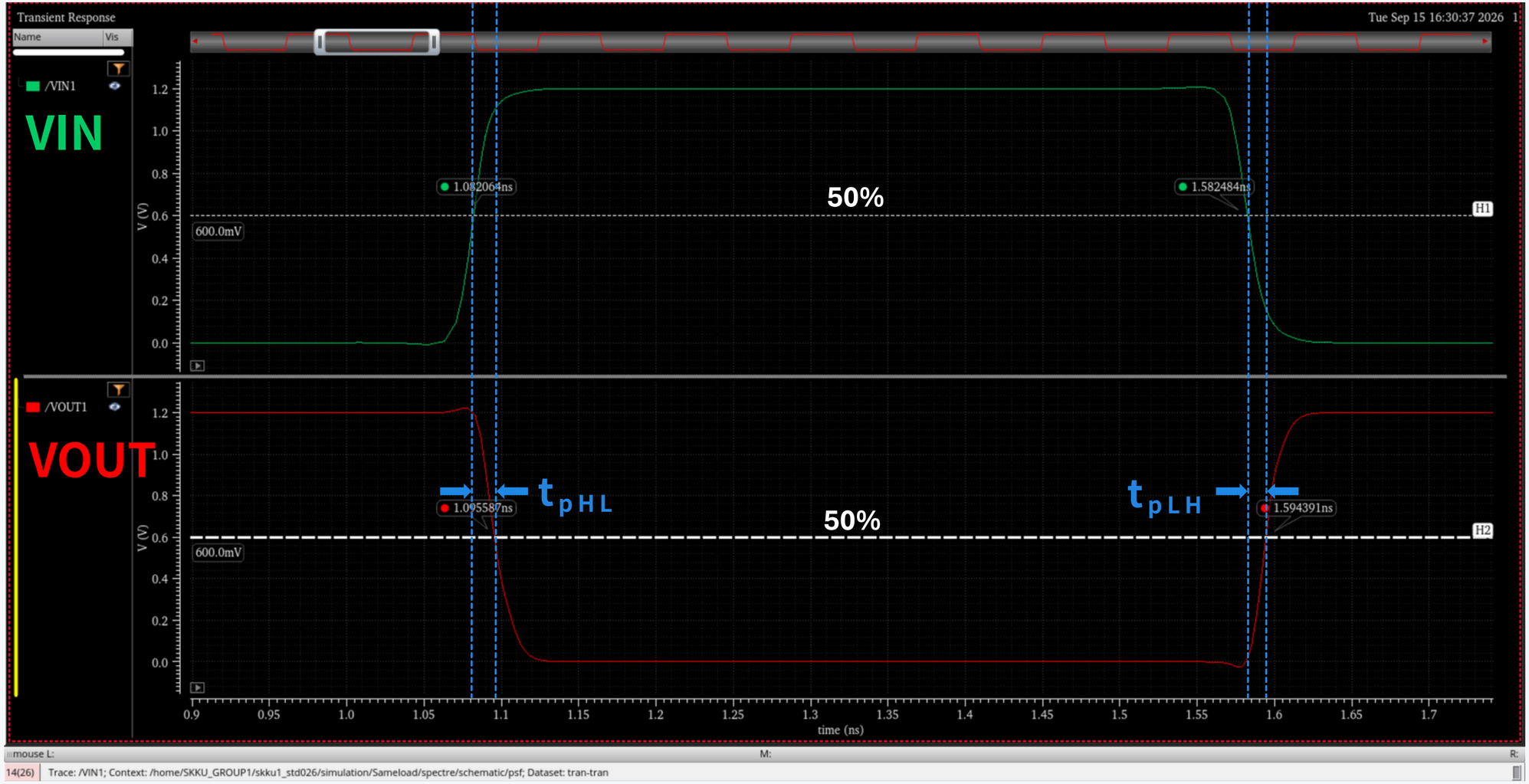
**$t_{pd}$**

대표적인 Propagation Delay

$t_{pd} = (t_{pHL} + t_{pLH}) / 2$

# Propagation Delay

## [ 90n process :Transient simulation result – VIN, VOUT ]



# Same Load

| Process Tech [nm] | Multiplier factor | Process Corner | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | SS(85°C) | | | NN(27°C) | | | FF(-25°C) | | |
| | | tpHL [ps] | tpLH [ps] | tpd [ps] | tpHL [ps] | tpLH [ps] | tpd [ps] | tpHL [ps] | tpLH [ps] | tpd [ps] |
| gpdk 180 | 1x | 41.07 | 51.86 | 46.47 | 30.92 | 38.51 | 34.72 | 23.98 | 29.77 | 26.88 |
| | 2x | 41.07 | 51.86 | 46.47 | 30.92 | 38.51 | 34.72 | 23.98 | 29.77 | 26.88 |
| | 4x | 41.07 | 51.86 | 46.47 | 30.92 | 38.51 | 34.72 | 23.98 | 29.77 | 26.88 |
| gpdk 090 | 1x | 23.61 | 19.87 | 21.74 | 13.46 | 11.96 | 12.71 | 8.71 | 8.36 | 8.535 |
| | 2x | 23.61 | 19.87 | 21.74 | 13.46 | 11.96 | 12.71 | 8.71 | 8.36 | 8.535 |
| | 4x | 23.61 | 19.87 | 21.74 | 13.46 | 11.96 | 12.71 | 8.71 | 8.36 | 8.535 |
| gpdk 045 | 1x | 12.00 | 12.13 | 12.07 | 7.84 | 8.23 | 8.04 | 5.26 | 5.87 | 5.57 |
| | 2x | 12.00 | 12.13 | 12.07 | 7.84 | 8.23 | 8.04 | 5.26 | 5.87 | 5.57 |
| | 4x | 12.00 | 12.13 | 12.07 | 7.84 | 8.23 | 8.04 | 5.26 | 5.87 | 5.57 |
| gpdk090 +2 inv. | 1x | 23.93 | 19.67 | 21.8 | 13.39 | 11.76 | 12.58 | 8.63 | 8.08 | 8.36 |
| | 2x | 23.93 | 19.67 | 21.8 | 13.39 | 11.76 | 12.58 | 8.63 | 8.08 | 8.36 |
| | 4x | 23.93 | 19.67 | 21.8 | 13.39 | 11.76 | 12.58 | 8.63 | 8.08 | 8.36 |

① Multiplier 변화에도 Delay 일정
$R_{on}$ ↓ ↔ $C_L$ ↑ → $R_{on} \cdot C_L$ ≈ 일정

② 공정 미세화
180 nm → 90 nm → 45 nm :
채널 길이 ↓ → 구동 전류 ↑
→ $R_{on}$ ↓, 기생 Capacitance ↓
→ $R_{on}C_L$ 감소 → Propagation Delay ↓

# Same Load

| Process Tech [nm] | Multiplier factor | Process Corner | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | SS(85°C) | | | NN(27°C) | | | FF(-25°C) | | |
| | | tpHL [ps] | tpLH [ps] | tpd [ps] | tpHL [ps] | tpLH [ps] | tpd [ps] | tpHL [ps] | tpLH [ps] | tpd [ps] |
| gpdk 180 | 1x | 41.07 | 51.86 | 46.47 | 30.92 | 38.51 | 34.72 | 23.98 | 29.77 | 26.88 |
| | 2x | 41.07 | 51.86 | 46.47 | 30.92 | 38.51 | 34.72 | 23.98 | 29.77 | 26.88 |
| | 4x | 41.07 | 51.86 | 46.47 | 30.92 | 38.51 | 34.72 | 23.98 | 29.77 | 26.88 |
| gpdk 090 | 1x | 23.61 | 19.87 | 21.74 | 13.46 | 11.96 | 12.71 | 8.71 | 8.36 | 8.535 |
| | 2x | 23.61 | 19.87 | 21.74 | 13.46 | 11.96 | 12.71 | 8.71 | 8.36 | 8.535 |
| | 4x | 23.61 | 19.87 | 21.74 | 13.46 | 11.96 | 12.71 | 8.71 | 8.36 | 8.535 |
| gpdk 045 | 1x | 12.00 | 12.13 | 12.07 | 7.84 | 8.23 | 8.04 | 5.26 | 5.87 | 5.57 |
| | 2x | 12.00 | 12.13 | 12.07 | 7.84 | 8.23 | 8.04 | 5.26 | 5.87 | 5.57 |
| | 4x | 12.00 | 12.13 | 12.07 | 7.84 | 8.23 | 8.04 | 5.26 | 5.87 | 5.57 |
| gpdk090 +2 inv. | 1x | 23.93 | 19.67 | 21.8 | 13.39 | 11.76 | 12.58 | 8.63 | 8.08 | 8.36 |
| | 2x | 23.93 | 19.67 | 21.8 | 13.39 | 11.76 | 12.58 | 8.63 | 8.08 | 8.36 |
| | 4x | 23.93 | 19.67 | 21.8 | 13.39 | 11.76 | 12.58 | 8.63 | 8.08 | 8.36 |

③ Process Corner 영향
SS → NN → FF :
트랜지스터 speed ↑ ( $R_{on} \approx V_{DD}/I_{ON}$ )
$I_{ON}$ ↑ → $R_{on}$ ↓ → Delay ↓

# Multi Load

| Process Tech [nm] | Multiplier factor | Process Corner | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | SS(85°C) | | | NN(27°C) | | | FF(-25°C) | | |
| | | tpHL [ps] | tpLH [ps] | tpd [ps] | tpHL [ps] | tpLH [ps] | tpd [ps] | tpHL [ps] | tpLH [ps] | tpd [ps] |
| gpdk 180 | 1x | 41.00 | 51.88 | 46.44 | 30.94 | 38.47 | 34.70 | 23.93 | 29.68 | 26.81 |
| | 2x | 53.88 | 66.00 | 59.94 | 40.90 | 50.01 | 45.46 | 31.68 | 39.14 | 35.41 |
| | 4x | 79.26 | 92.25 | 85.75 | 60.65 | 71.71 | 66.18 | 47.15 | 57.12 | 52.14 |
| gpdk 090 | 1x | 23.6 | 19.86 | 21.73 | 13.47 | 11.96 | 12.715 | 8.68 | 8.36 | 8.52 |
| | 2x | 32.88 | 27.49 | 30.185 | 18.98 | 16.65 | 17.815 | 12.39 | 11.67 | 12.03 |
| | 4x | 50.75 | 42.07 | 46.41 | 29.77 | 25.7 | 27.735 | 19.64 | 18.27 | 18.955 |
| gpdk 045 | 1x | 12.00 | 12.14 | 12.07 | 7.83 | 8.23 | 8.03 | 5.25 | 5.88 | 5.56 |
| | 2x | 14.56 | 14.72 | 14.64 | 9.57 | 10.34 | 9.96 | 6.55 | 7.59 | 7.07 |
| | 4x | 19.18 | 19.98 | 19.58 | 12.85 | 14.47 | 13.66 | 9.02 | 10.95 | 9.98 |
| gpdk090 +2 inv. | 1x | 23.93 | 19.67 | 21.8 | 13.39 | 11.76 | 12.58 | 8.63 | 8.08 | 8.36 |
| | 2x | 33.68 | 27.00 | 30.34 | 19.01 | 16.37 | 17.69 | 12.35 | 11.40 | 11.88 |
| | 4x | 53.42 | 40.50 | 49.96 | 30.16 | 25.18 | 27.67 | 19.65 | 18.00 | 18.83 |

## 1) 공정 미세화

180 nm → 90 nm → 45 nm :
채널 길이 ↓ → 구동 전류 ↑
→ $R_{on}$ ↓, 기생 Capacitance ↓
→ $R_{on}C_L$ 감소 → Propagation Delay ↓

# Multi Load

| Process Tech [nm] | Multiplier factor | Process Corner | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | SS(85°C) | | | NN(27°C) | | | FF(-25°C) | | |
| | | tpHL [ps] | tpLH [ps] | tpd [ps] | tpHL [ps] | tpLH [ps] | tpd [ps] | tpHL [ps] | tpLH [ps] | tpd [ps] |
| gpdk 180 | 1x | 41.00 | 51.88 | 46.44 | 30.94 | 38.47 | 34.70 | 23.93 | 29.68 | 26.81 |
| | 2x | 53.88 | 66.00 | 59.94 | 40.90 | 50.01 | 45.46 | 31.68 | 39.14 | 35.41 |
| | 4x | 79.26 | 92.25 | 85.75 | 60.65 | 71.71 | 66.18 | 47.15 | 57.12 | 52.14 |
| gpdk 090 | 1x | 23.6 | 19.86 | 21.73 | 13.47 | 11.96 | 12.715 | 8.68 | 8.36 | 8.52 |
| | 2x | 32.88 | 27.49 | 30.185 | 18.98 | 16.65 | 17.815 | 12.39 | 11.67 | 12.03 |
| | 4x | 50.75 | 42.07 | 46.41 | 29.77 | 25.7 | 27.735 | 19.64 | 18.27 | 18.955 |
| gpdk 045 | 1x | 12.00 | 12.14 | 12.07 | 7.83 | 8.23 | 8.03 | 5.25 | 5.88 | 5.56 |
| | 2x | 14.56 | 14.72 | 14.64 | 9.57 | 10.34 | 9.96 | 6.55 | 7.59 | 7.07 |
| | 4x | 19.18 | 19.98 | 19.58 | 12.85 | 14.47 | 13.66 | 9.02 | 10.95 | 9.98 |
| gpdk090 +2 inv. | 1x | 23.93 | 19.67 | 21.8 | 13.39 | 11.76 | 12.58 | 8.63 | 8.08 | 8.36 |
| | 2x | 33.68 | 27.00 | 30.34 | 19.01 | 16.37 | 17.69 | 12.35 | 11.40 | 11.88 |
| | 4x | 53.42 | 40.50 | 49.96 | 30.16 | 25.18 | 27.67 | 19.65 | 18.00 | 18.83 |

## 2) Multiplier 증가에 따른 Delay 증가

$$t_{pd} \approx 0.69 \cdot R_{on} \cdot C_L$$

Multiplier ↑

$R_{on}$ ↓ : R → R/2 → R/4

$C_L$ ↑ : $1C_L$ → $3C_L$ → $10C_L$

이론상 tpd : 1배 → 1.5배 → 2.5배

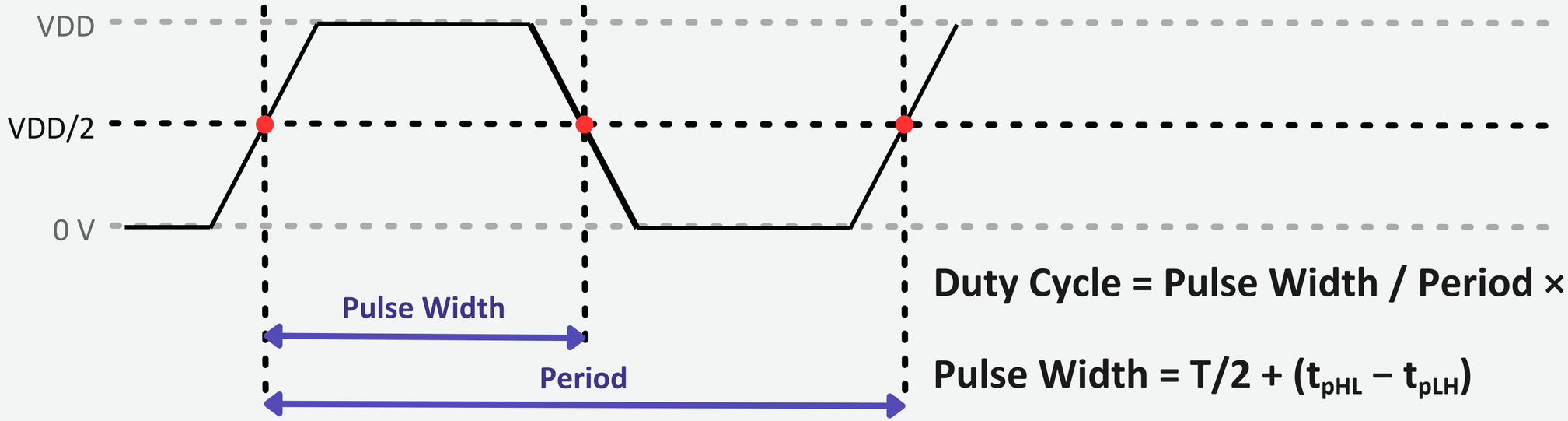
실측 tpd : 1배 → 1.43배 → 2.23배

원인 : tpd는 이상적인 1차 RC 모델

# Duty Cycle

# Duty Cycle

- 한 주기 T 동안 출력이 HIGH로 유지되는 시간의 비율
- Clock 파형에서 50% 교차점을 이용해 Pulse Width와 Period를 측정




**Duty Cycle = Pulse Width / Period × 100 %**

**Pulse Width = T/2 + ($t_{pHL}$ – $t_{pLH}$)**

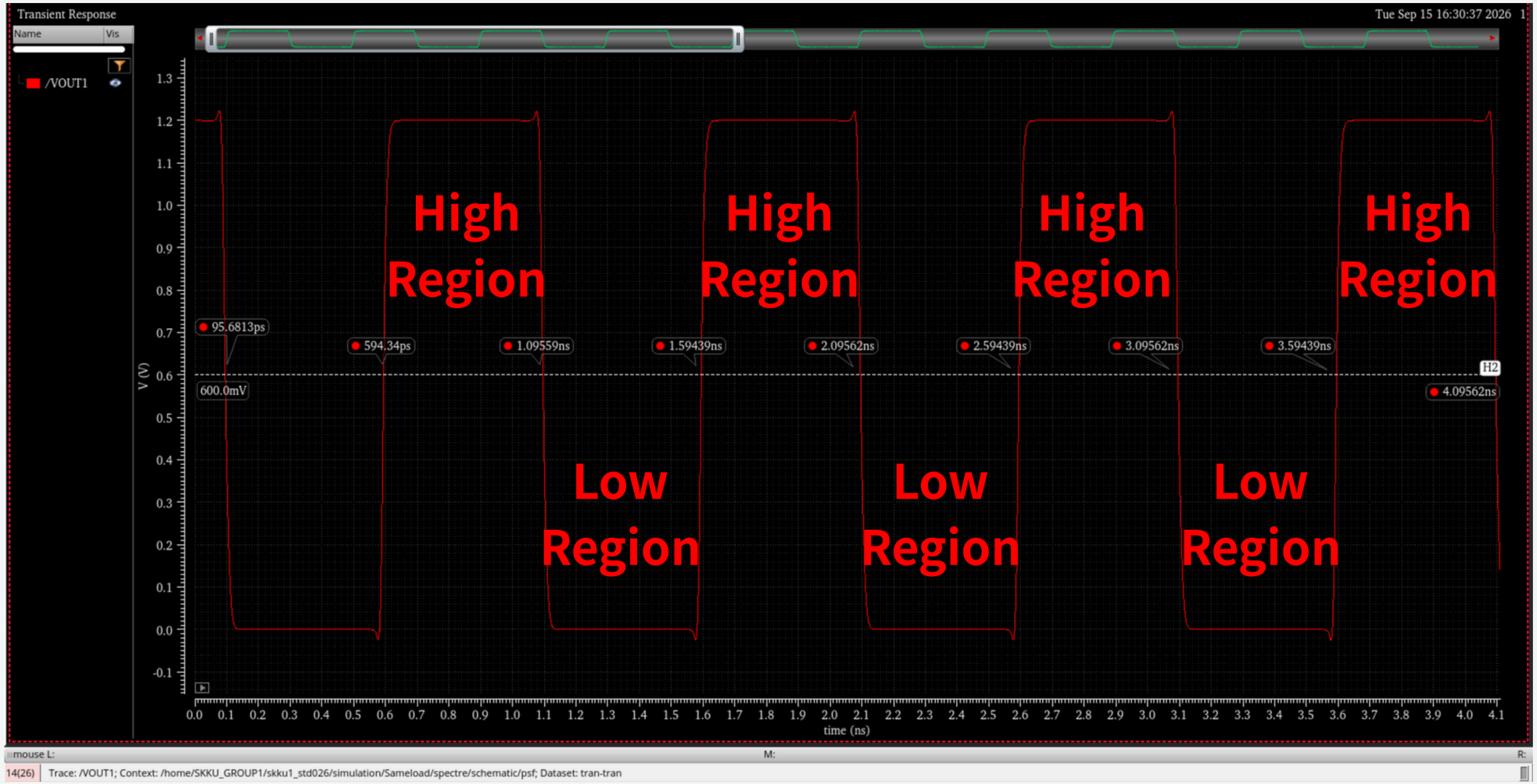
$$t_{High} = \frac{T}{2} + (t_{pHL} - t_{pLH}) \quad Duty\ High(\%) = \frac{t_{High}}{T} \times 100$$

**핵심 관계**

$t_{pHL} \approx 0.69RnCL$ / $t_{pLH} \approx 0.69RpCL$

**→ Duty Cycle은 절대적인 delay보다 $t_{pHL}$과 $t_{pLH}$의 상대적인 차이에 민감**

# Duty Cycle


Transient Response
Tue Sep 15 16:30:37 2026
Name
Vis
/VOUT1
V (V)
time (ns)
High Region
High Region
High Region
High Region
Low Region
Low Region
Low Region
95.6813ps
594.34ps
1.09559ns
1.59439ns
2.09562ns
2.59439ns
3.09562ns
3.59439ns
4.09562ns
600.0mV
H2
mouse L:
M:
R:
14(26)
Trace: /VOUT1; Context: /home/SKKU_GROUP1/skku1_std026/simulation/Sameload/spectre/schematic/psf; Dataset: tran-tran

# Same Load

Pulse Width = T/2 + ($t_{pHL}$ – $t_{pLH}$)

| Process Tech [nm] | Multiplier factor | Process Corner | | |
|---|---|---|---|---|
| | | SS | NN | FF |
| | | Duty H [%] | Duty H [%] | Duty H [%] |
| gpdk 180 | 1x | 51.08 | 50.62 | 50.37 |
| | 2x | 51.08 | 50.62 | 50.37 |
| | 4x | 51.08 | 50.62 | 50.37 |
| gpdk 090 | 1x | 50.63 | 50.12 | 49.77 |
| | 2x | 50.63 | 50.12 | 49.77 |
| | 4x | 50.63 | 50.12 | 49.77 |
| gpdk 045 | 1x | 50.08 | 49.93 | 49.86 |
| | 2x | 50.08 | 49.93 | 49.86 |
| | 4x | 50.08 | 49.93 | 49.86 |
| gpdk090 +2 inv. | 1x | 50.57 | 50.12 | 49.78 |
| | 2x | 50.57 | 50.12 | 49.78 |
| | 4x | 50.57 | 50.12 | 49.78 |

### 1. SS > NN > FF

- SS : $t_{pHL} > t_{pLH}$
  NMOS의 방전 지연과 PMOS의 충전 지연 사이의 상대적인 차이
- PVT variation의 비대칭적 영향

### 2. 공정 미세화될수록 duty 50%에 가까워진다

- 미세공정일수록 velocity saturation의 영향이 커져 딜레이 차이 줄어듦
- raising/falling 특성 더 대칭적으로 변함

### 3. Multiply factor는 duty에 영향이 없다

- Multiplier factor 는 NMOS/PMOS 폭을 같은 비율로 키움(비대칭성 유지)
- duty는 공정 특성에 의해 결정

# Multi Load

| Process Tech [nm] | Multiplier factor | Process Corner | | |
|---|---|---|---|---|
| | | SS | NN | FF |
| | | Duty H [%] | Duty H [%] | Duty H [%] |
| gpdk 180 | 1x | 51.08 | 50.62 | 50.37 |
| | 2x | 51.22 | 50.78 | 50.55 |
| | 4x | 51.31 | 50.98 | 50.80 |
| gpdk 090 | 1x | 50.63 | 50.13 | 49.77 |
| | 2x | 50.89 | 50.23 | 49.81 |
| | 4x | 51.49 | 50.46 | 49.88 |
| gpdk 045 | 1x | 50.08 | 49.93 | 49.86 |
| | 2x | 50.05 | 49.89 | 49.82 |
| | 4x | 49.98 | 49.81 | 49.73 |
| gpdk090 +2 inv. | 1x | 50.57 | 50.12 | 49.78 |
| | 2x | 50.75 | 50.21 | 49.82 |
| | 4x | 51.08 | 50.39 | 49.89 |

### 1. SS > NN > FF

- Same Load 와 동일하게 성립함

### 2. Multiplier factor 커질수록 duty 값이 커진다

- 부하가 커지면 driver sizing과 부하 사이에 상호 작용이 duty 비대칭을 키움

### 3. 크기를 키울수록 50%에 가까워진다

- velocity saturation / short-channel effect 가 사이징 효과보다 우세하게 작용

# Power Consumption

# Same Load

| Process Tech [nm] | Multiplier factor | Process Corner | | | | | |
|---|---|---|---|---|---|---|---|
| | | SS(85°C) | | NN(27°C) | | FF(-25°C) | |
| | | IDD [uA] | ISS [uA] | IDD [uA] | ISS [uA] | IDD [uA] | ISS [uA] |
| gpdk 180 | 1x | 18.92 | 18.91 | 18.966 | 18.96 | 18.89 | 18.92 |
| | 2x | 37.84 | 37.83 | 37.93 | 37.91 | 37.78 | 37.84 |
| | 4x | 75.69 | 75.65 | 75.86 | 75.82 | 75.57 | 75.67 |
| gpdk 090 | 1x | 2.12 | 2.13 | 2.18 | 2.18 | 2.24 | 2.24 |
| | 2x | 4.26 | 4.25 | 4.37 | 4.35 | 4.49 | 4.49 |
| | 4x | 8.51 | 8.51 | 8.73 | 8.70 | 8.98 | 8.97 |
| gpdk 045 | 1x | 0.45 | 0.45 | 0.47 | 0.47 | 0.49 | 0.49 |
| | 2x | 0.91 | 0.91 | 0.94 | 0.94 | 0.98 | 0.98 |
| | 4x | 1.82 | 1.82 | 1.89 | 1.89 | 1.96 | 1.96 |
| gpdk090 +2 inv. | 1x | 2.13 | 2.12 | 2.15 | 2.15 | 2.13 | 2.13 |
| | 2x | 4.25 | 4.25 | 4.30 | 4.30 | 4.25 | 4.25 |
| | 4x | 8.50 | 8.50 | 8.62 | 8.61 | 8.50 | 8.50 |

# Same Load

| Process Tech [nm] | Multiplier factor | Process Corner | |
|---|---|---|---|
| | | NN(27°C) | |
| | | IDD [uA] | ISS [uA] |
| gpdk 180 | 1x | 18.966 | 18.96 |
| | 2x | 37.93 | 37.91 |
| | 4x | 75.86 | 75.82 |
| gpdk 090 | 1x | 2.18 | 2.18 |
| | 2x | 4.37 | 4.35 |
| | 4x | 8.73 | 8.70 |
| gpdk 045 | 1x | 0.47 | 0.47 |
| | 2x | 0.94 | 0.94 |
| | 4x | 1.89 | 1.89 |
| gpdk090 +2 inv. | 1x | 2.15 | 2.15 |
| | 2x | 4.30 | 4.30 |
| | 4x | 8.62 | 8.61 |

## 1) Process Tech 따른 분석

$P_{total}=\alpha f_{switch} C_L VDD^2$

$C_L=WL \times C_{ox}$

$P_{total} \propto W \times L$

**Length에 따른 power 감소**

## 2) Multiplier에 따른 분석

$R_{ON}$ → 1/N배 감소

$C_L$ → N배 증가

IDD, ISS → N배 증가

Power Consumption → N배 증가

# Multi Load

| Process Tech [nm] | Multiplier factor | Process Corner | | | | | |
|---|---|---|---|---|---|---|---|
| | | SS(85°C) | | NN(27°C) | | FF(-25°C) | |
| | | IDD [uA] | ISS [uA] | IDD [uA] | ISS [uA] | IDD [uA] | ISS [uA] |
| gpdk 180 | 1x | 18.91 | 18.91 | 18.99 | 18.97 | 18.95 | 18.90 |
| | 2x | 56.04 | 56.00 | 56.65 | 56.61 | 56.55 | 56.31 |
| | 4x | 184.55 | 184.42 | 187.84 | 187.73 | 187.26 | 187.00 |
| gpdk 090 | 1x | 2.13 | 2.12 | 2.18 | 2.18 | 2.25 | 2.24 |
| | 2x | 6.68 | 6.69 | 6.84 | 6.84 | 7.04 | 7.04 |
| | 4x | 23.11 | 23.10 | 23.67 | 23.65 | 24.31 | 24.30 |
| gpdk 045 | 1x | 0.45 | 0.45 | 0.47 | 0.47 | 0.49 | 0.49 |
| | 2x | 1.35 | 1.35 | 1.44 | 1.44 | 1.53 | 1.53 |
| | 4x | 4.45 | 4.45 | 4.83 | 4.83 | 5.22 | 5.23 |
| gpdk090 +2 inv. | 1x | 2.13 | 2.12 | 2.15 | 2.15 | 2.13 | 2.13 |
| | 2x | 6.69 | 6.69 | 6.80 | 6.81 | 6.97 | 6.97 |
| | 4x | 23.12 | 23.12 | 23.60 | 23.63 | 24.26 | 24.26 |

# Multi Load

| Process Tech [nm] | Multiplier factor | Process Corner | |
|---|---|---|---|
| | | NN(27°C) | |
| | | IDD [uA] | ISS [uA] |
| gpdk 180 | 1x | 18.99 | 18.97 |
| | 2x | 56.65 | 56.61 |
| | 4x | 187.84 | 187.73 |
| gpdk 090 | 1x | 2.18 | 2.18 |
| | 2x | 6.84 | 6.84 |
| | 4x | 23.67 | 23.65 |
| gpdk 045 | 1x | 0.47 | 0.47 |
| | 2x | 1.44 | 1.44 |
| | 4x | 4.83 | 4.83 |
| gpdk090 +2 inv. | 1x | 2.15 | 2.15 |
| | 2x | 6.80 | 6.81 |
| | 4x | 23.60 | 23.63 |

## 1) Process Tech 따른 분석

$P_{total}=\alpha f_{switch} C_L VDD^2$

$C_L=WL \times C_{ox}$

$P_{total} \propto W \times L$

**Length에 따른 power 감소**

## 2) Multiplier에 따른 분석

$R_{ON}$ → 1/N배 감소

$C_L$ → N배 증가

IDD, ISS → $(N+N^2)/2$배 증가

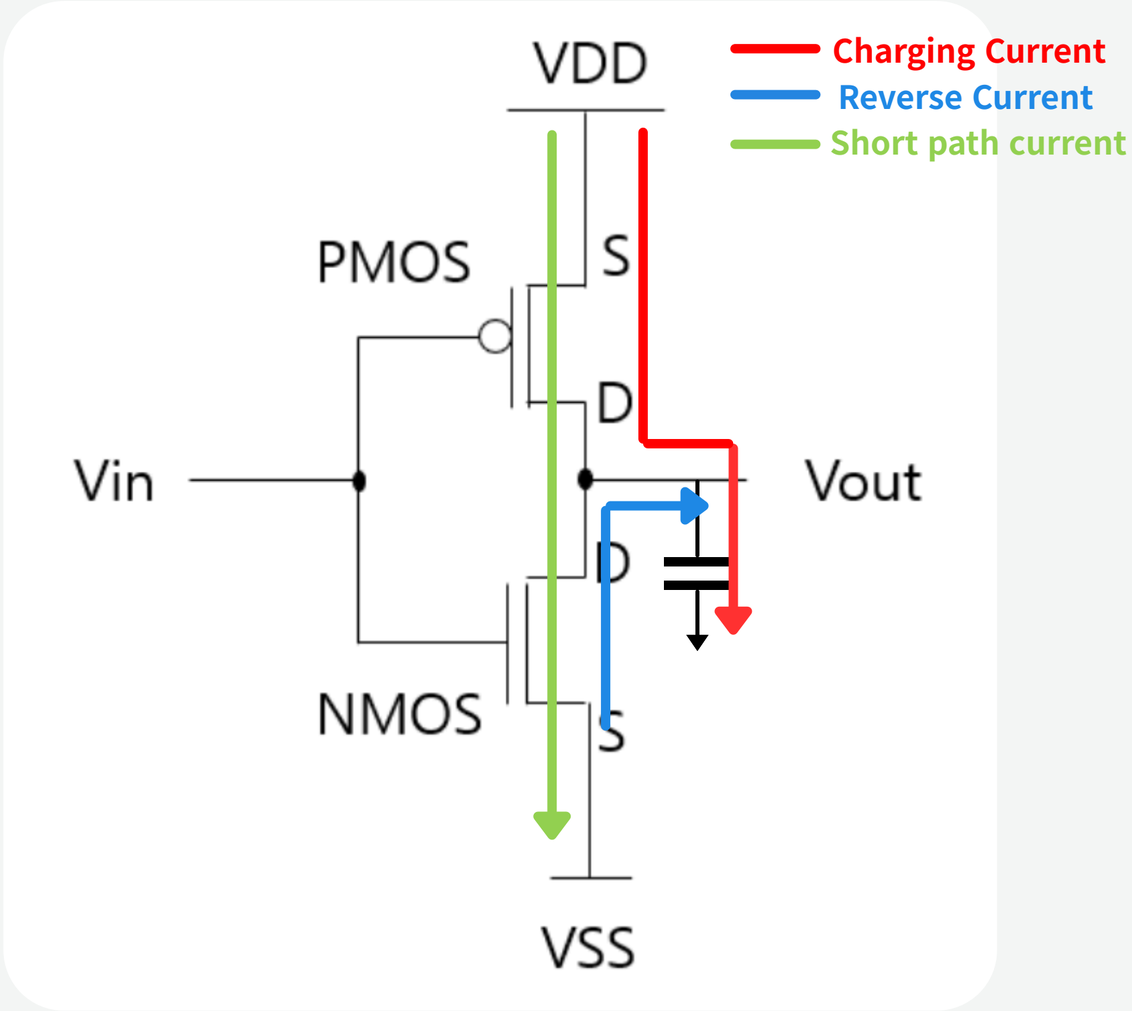
Power Consumption

→ $(N+N^2)/2$배 증가

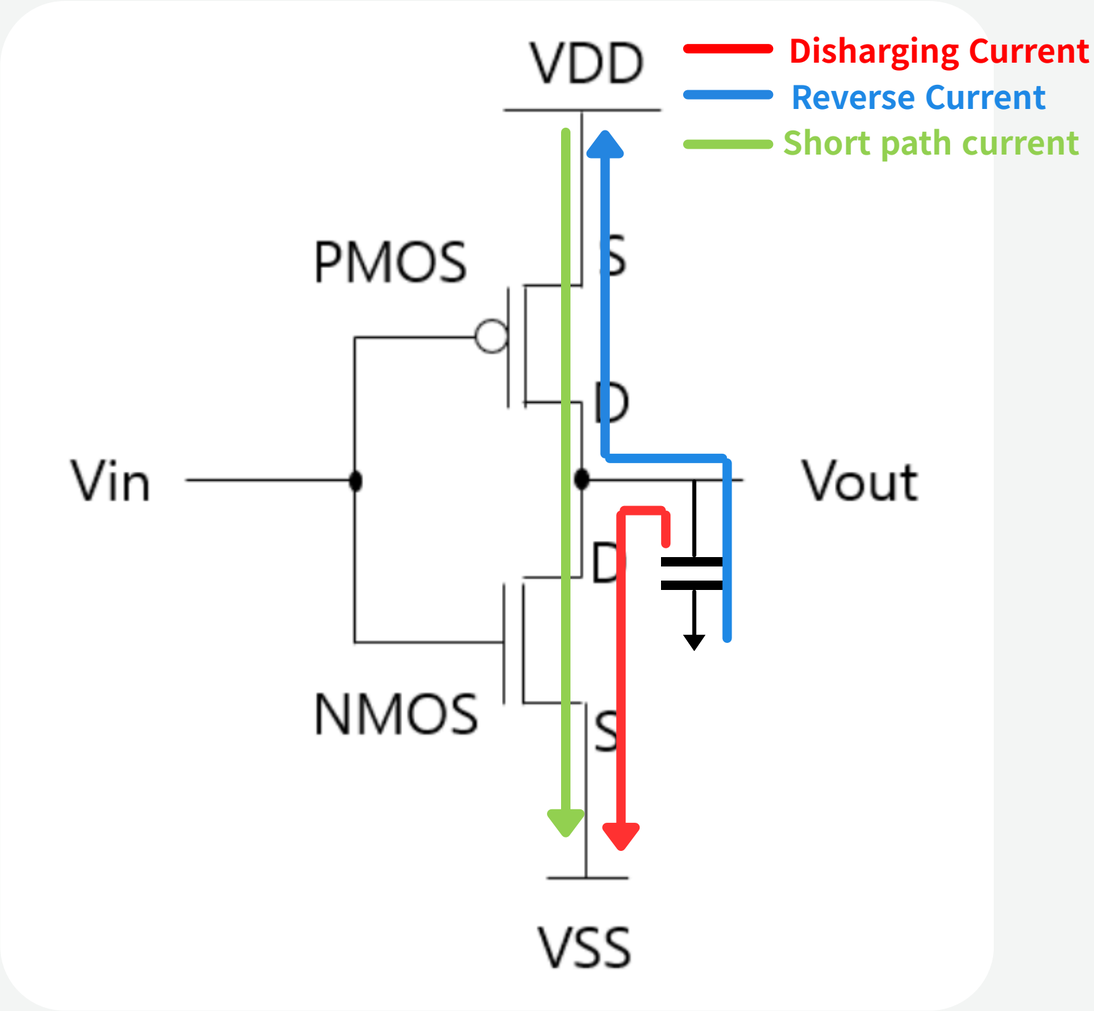
# Current Analysis

# Power Consumption - Current Analysis

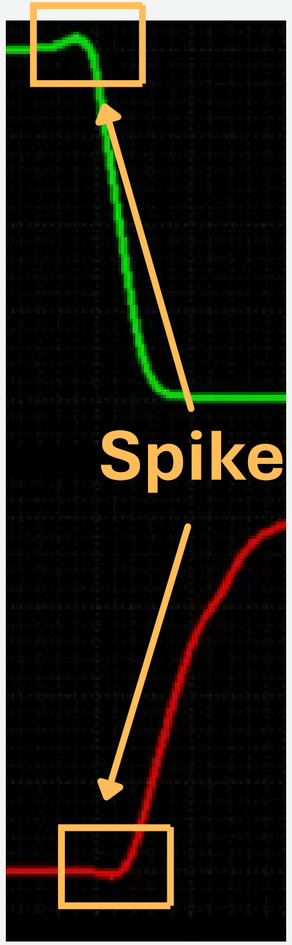
- **Charging Current**




- **Discharging Current**



# Current Analysis - Miller effect and capacitive coupling (Spike)




$$i_{Cgd} = C_{gd}\frac{d(V_{in} - V_{out})}{dt}$$

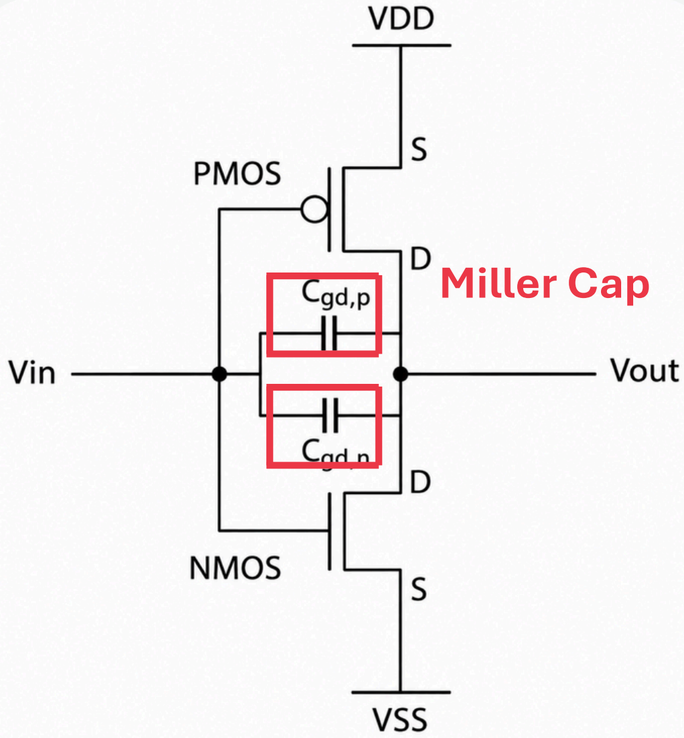
$$C_{Miller} \approx C_{gd}(1 - A_v)$$

Switching 순간 변화량 V가 급격히 변화

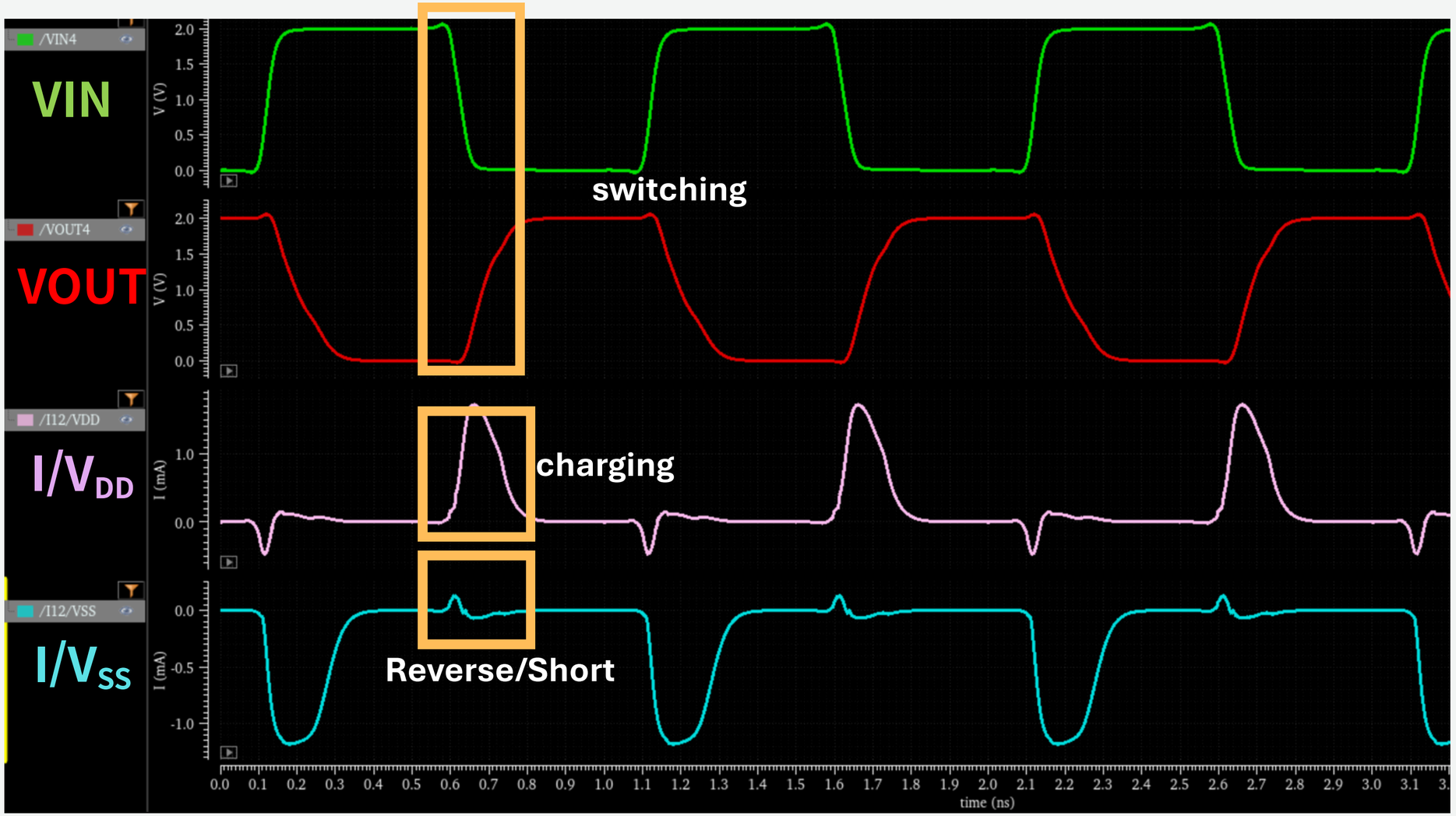
$C_{gd}$(Miller cap) 통해 순간적인 Capacitive Coupling Current 발생

M ↑ → $C_{gd,total}$ ↑ → Coupling Current ↑ → Spike 증가

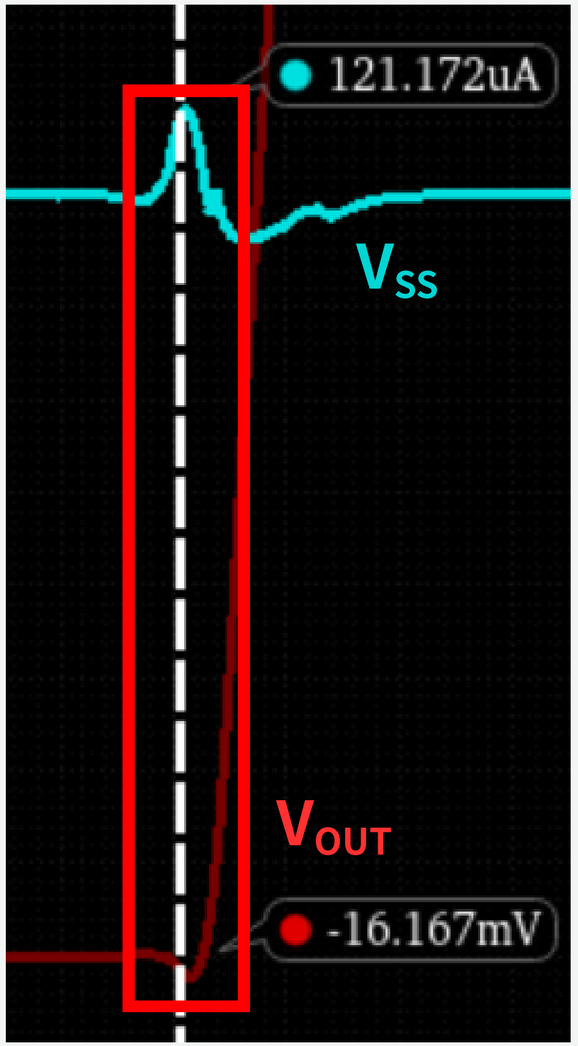
- **MOSFET M값 클수록 → Spike ↑**

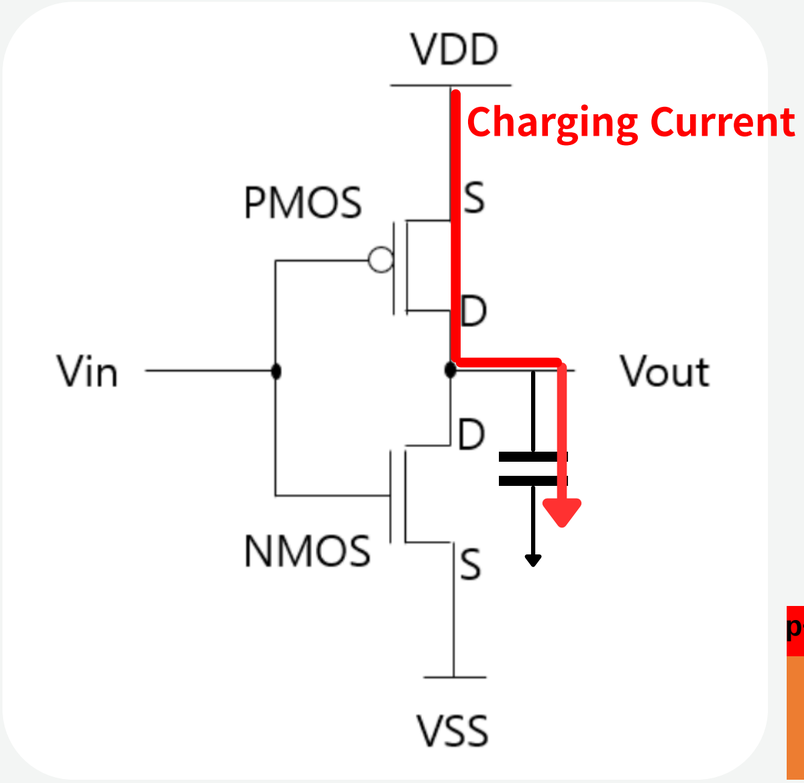


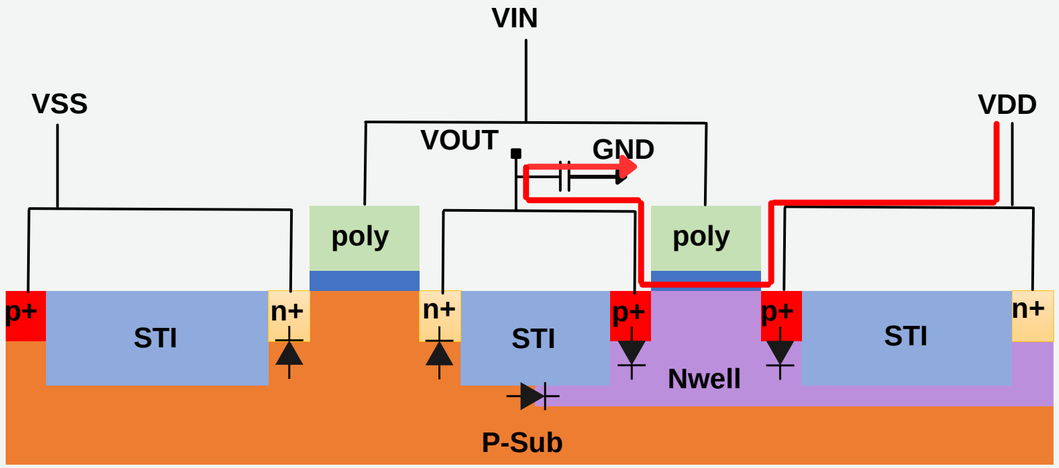
# Current Analysis - Charging Current



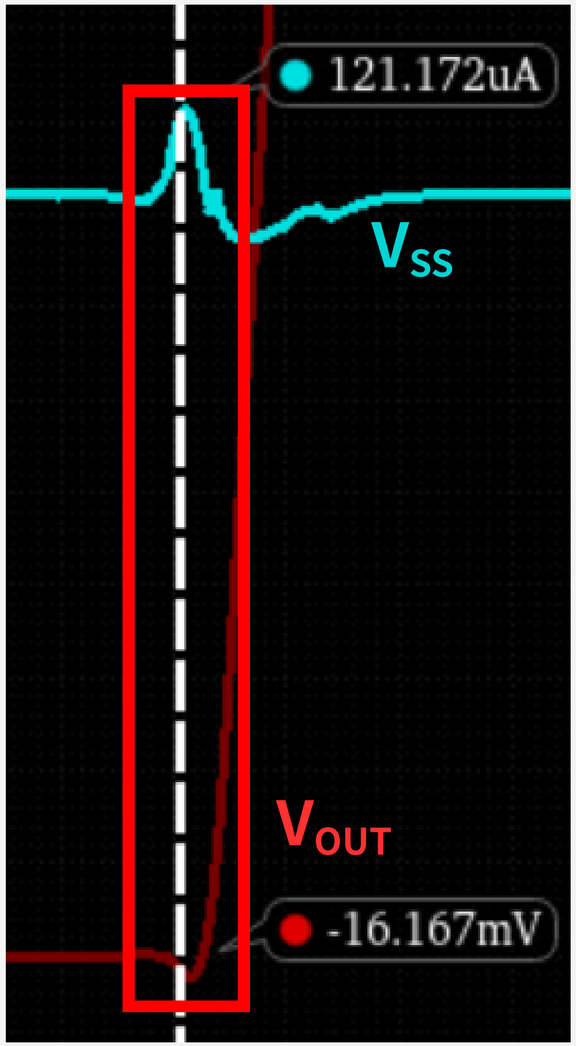
# Current Analysis - Charge Current



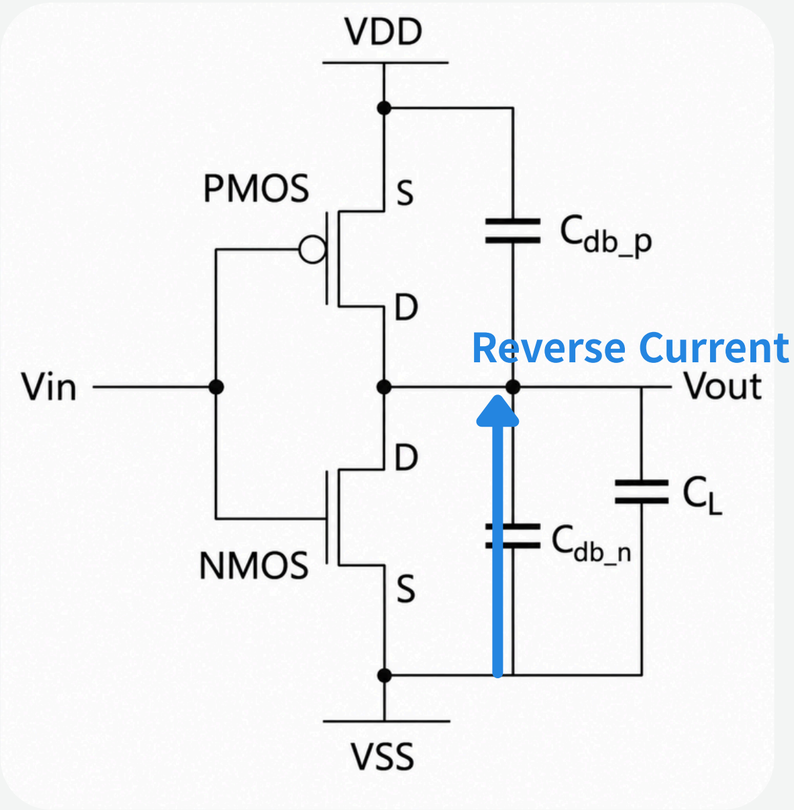



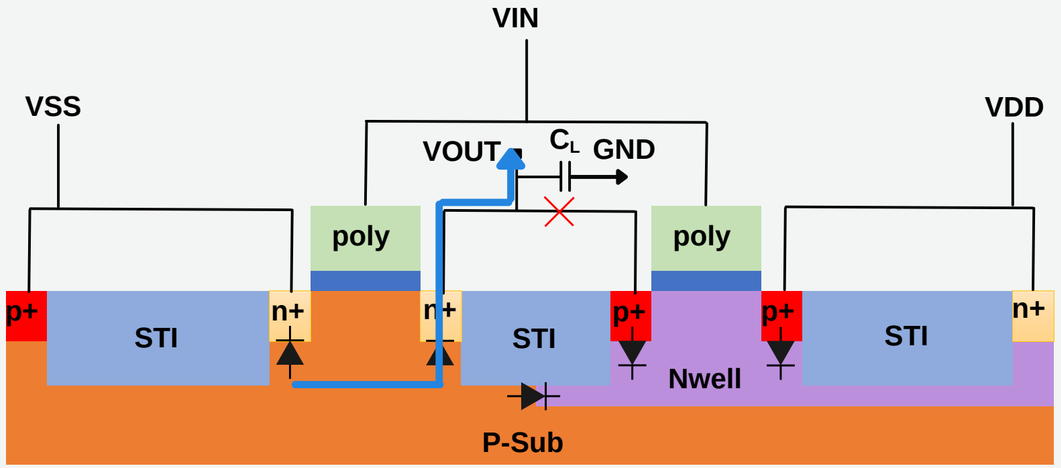



# Current Analysis - Charge Reverse Current



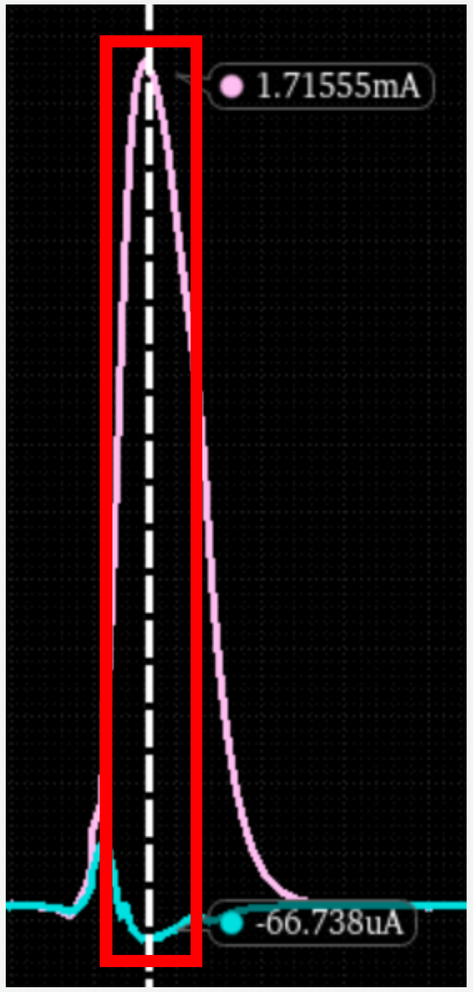

**$V_{out}$ Spike**
**→ Reverse current maximized**



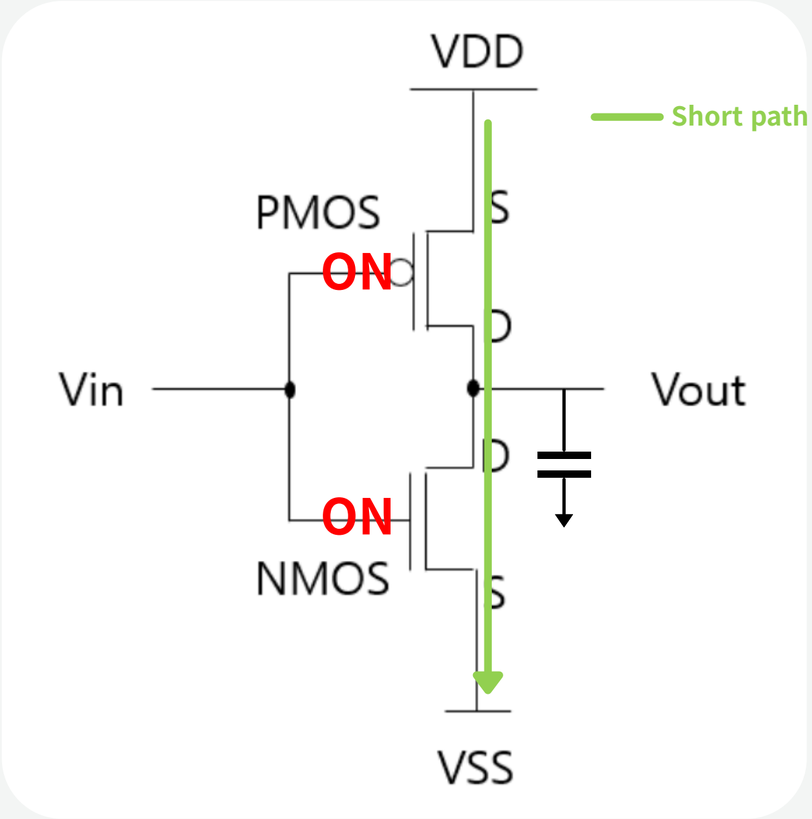



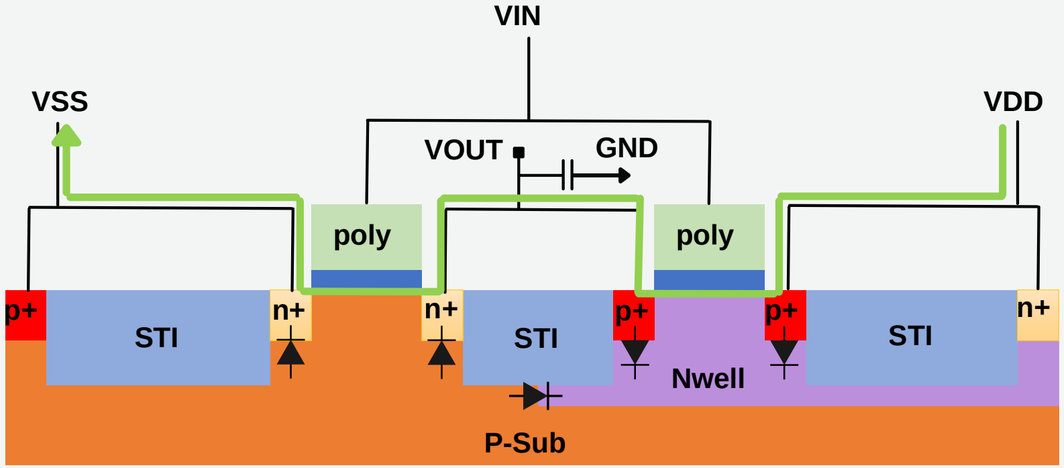
# Current Analysis - Charge Short Current

## Charging Current



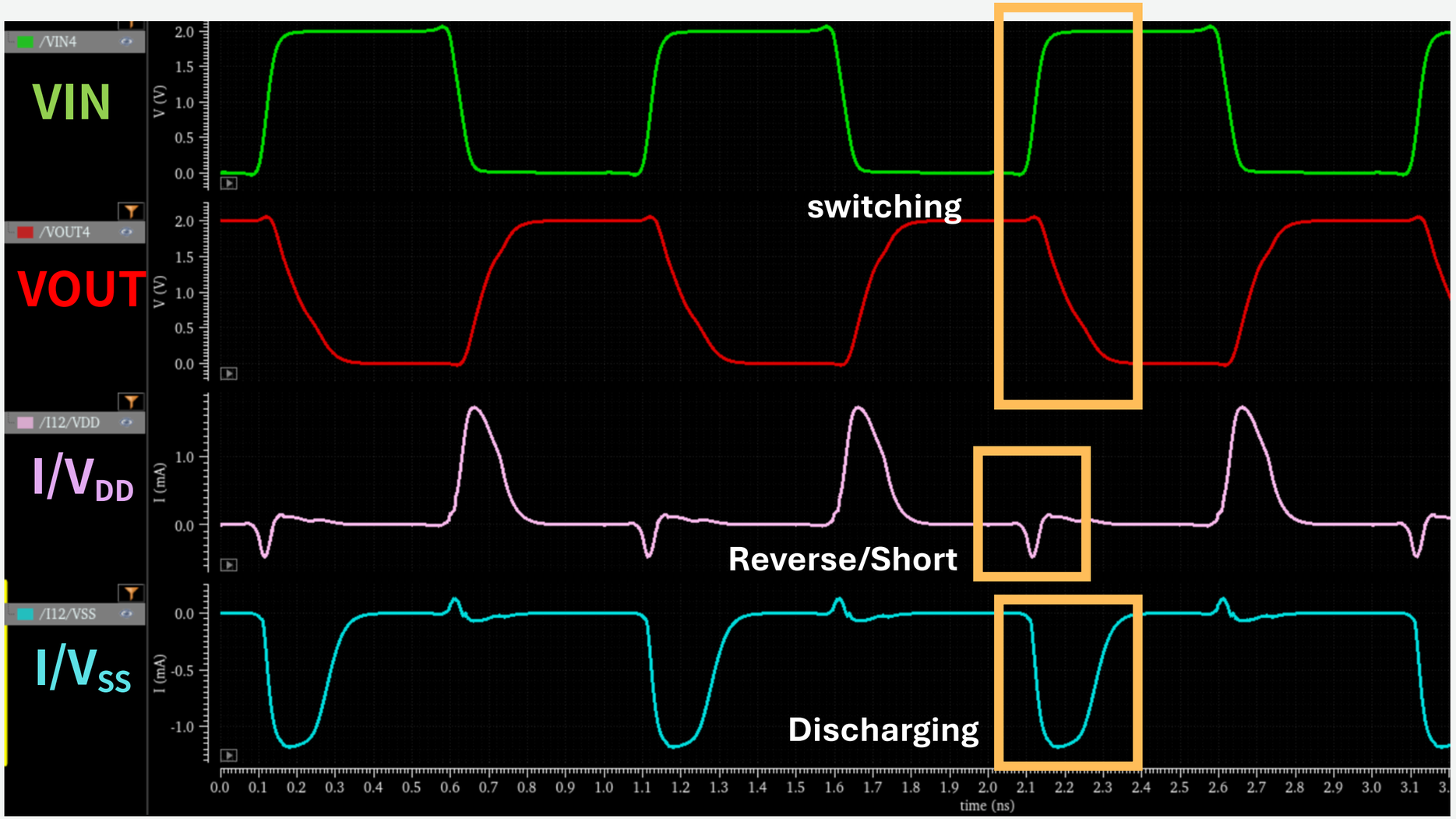

**Charge current is miximized, when short circuit is opened**



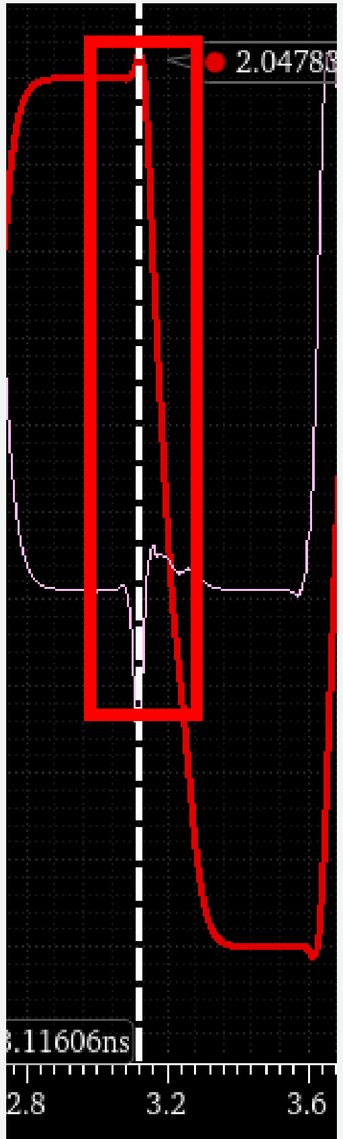




## Shot Path Current

# Current Analysis - Discharging Current

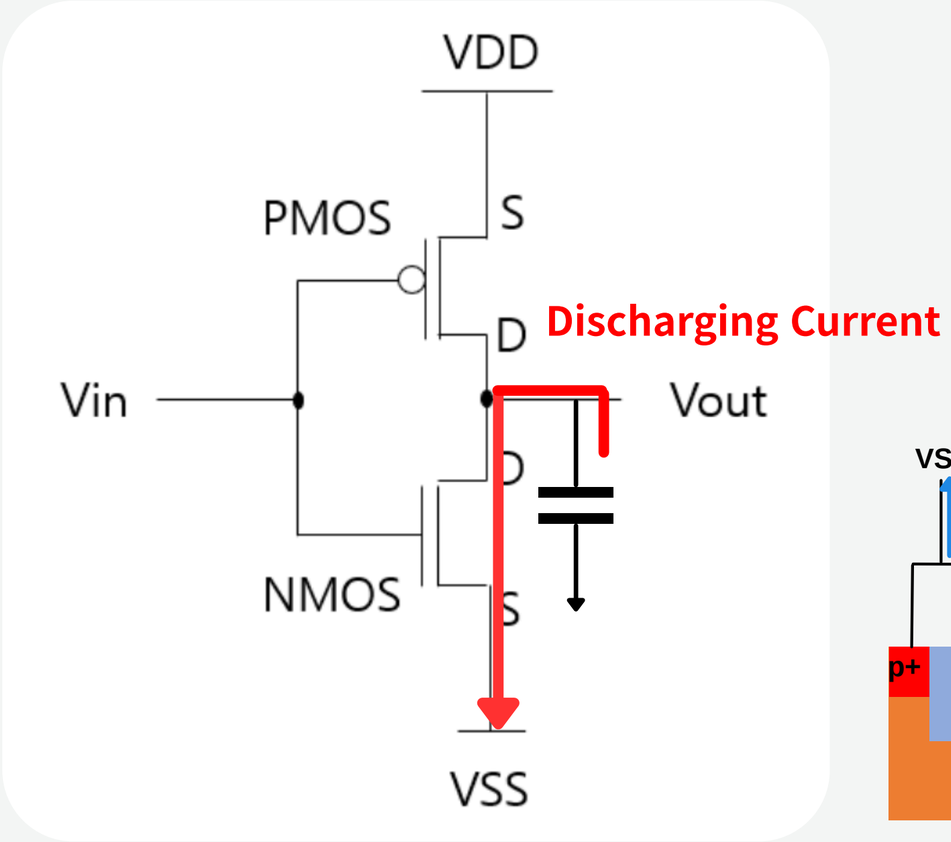


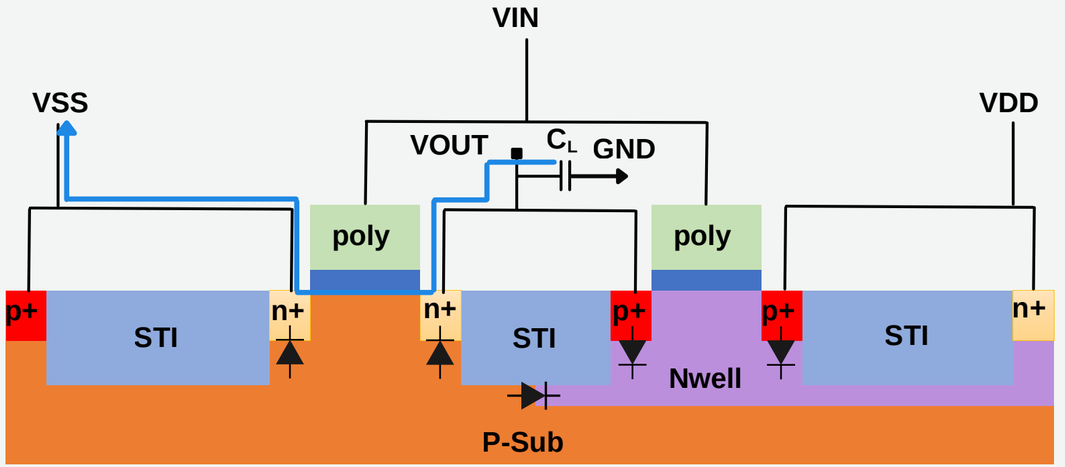
# Current Analysis - Discharge Current



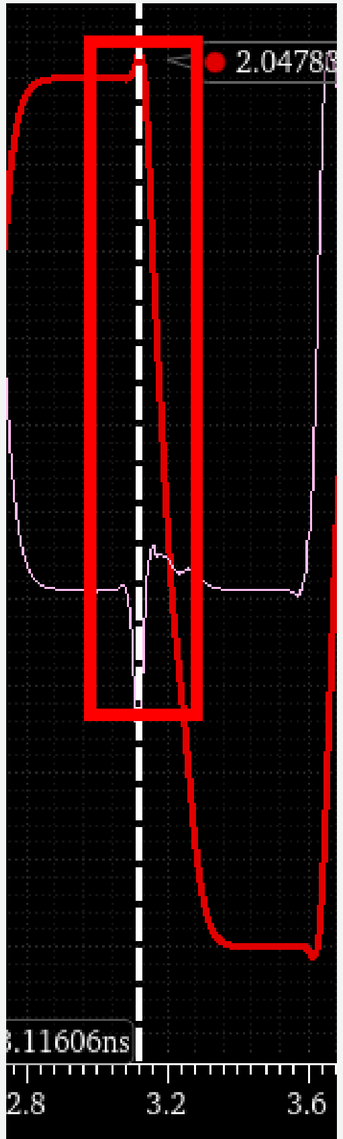

## $V_{out}$ Spike

## → Reverse current maximized



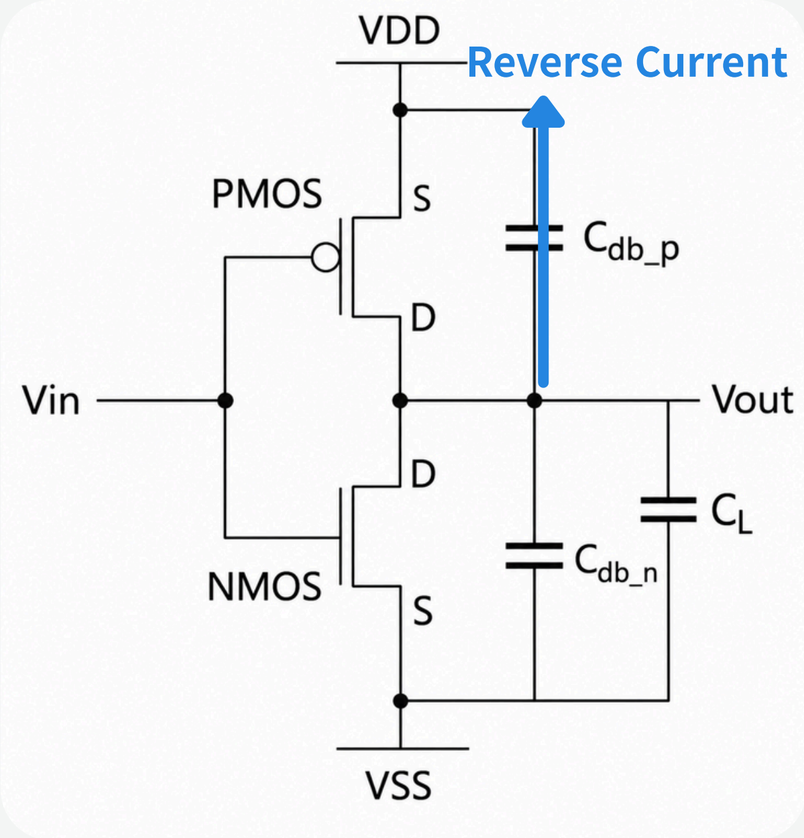



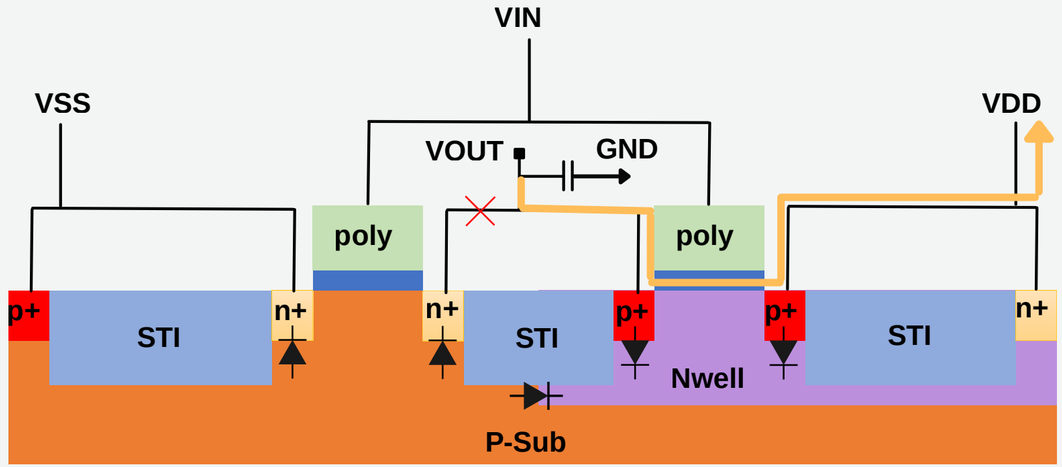
# Current Analysis - Discharge Reverse Current




## $V_{out}$ Spike

## → Reverse current maximized

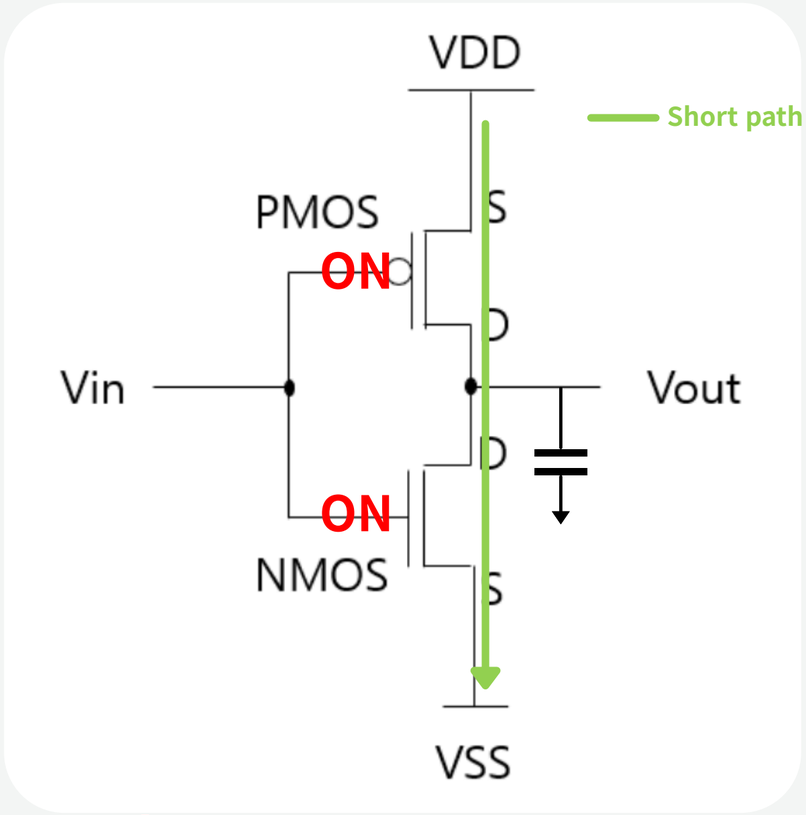


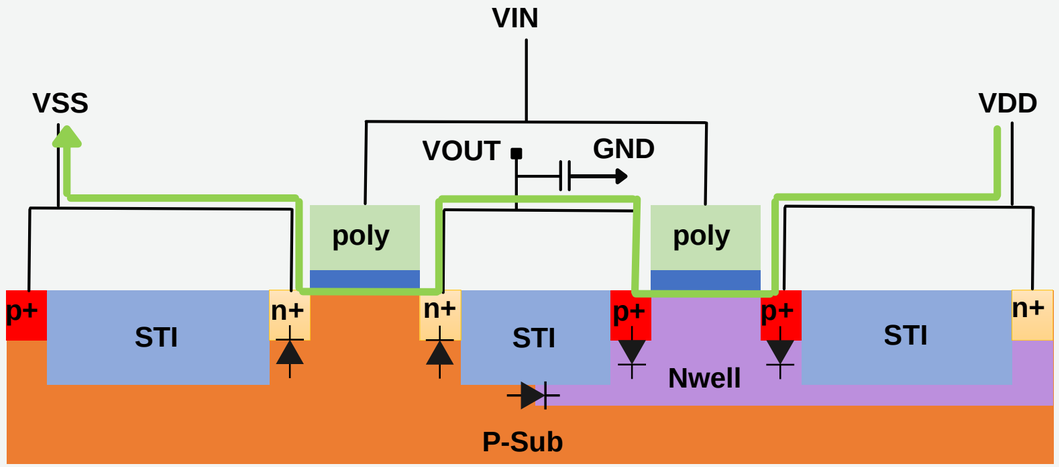



# Current Analysis - Discharge Short Path Current

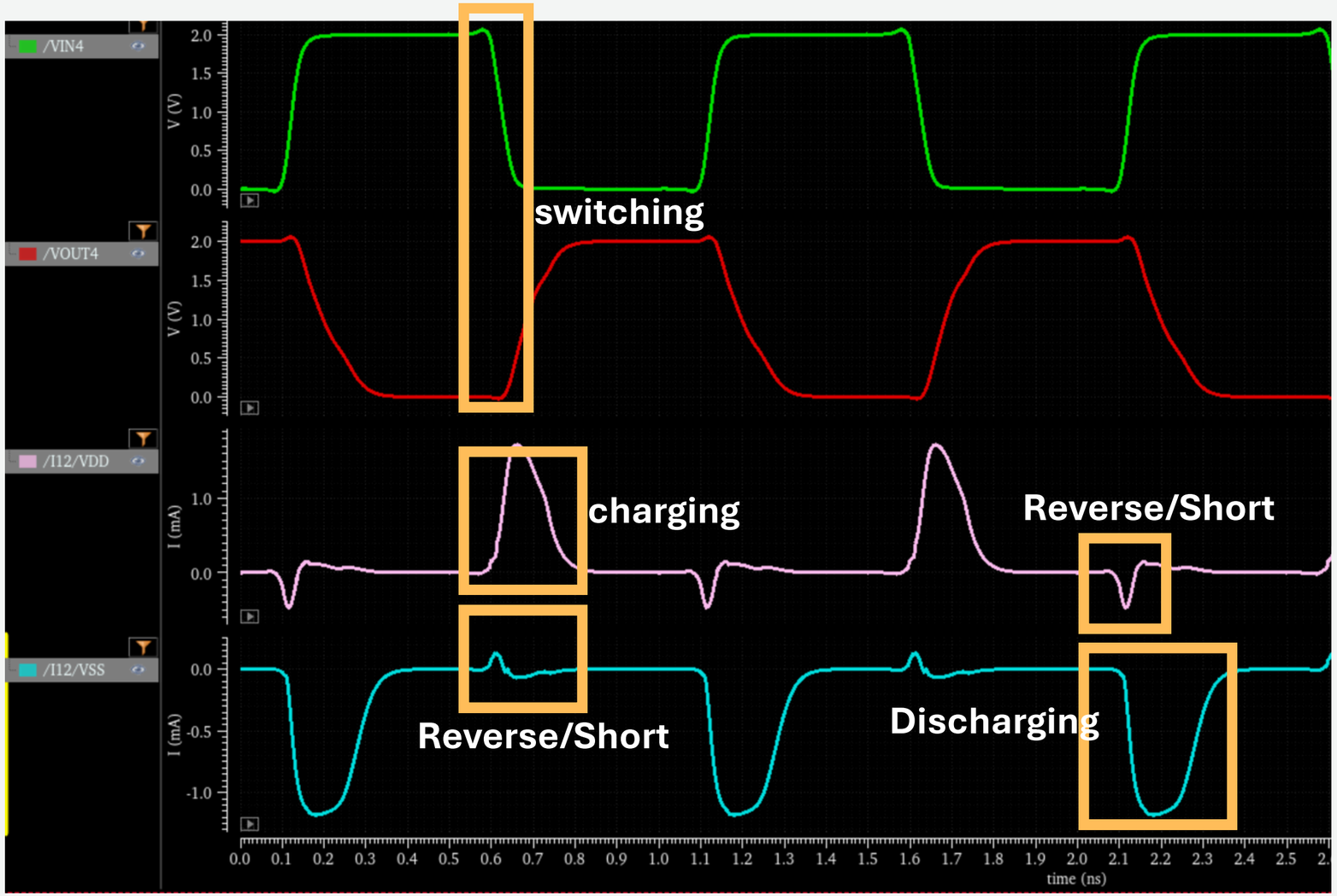


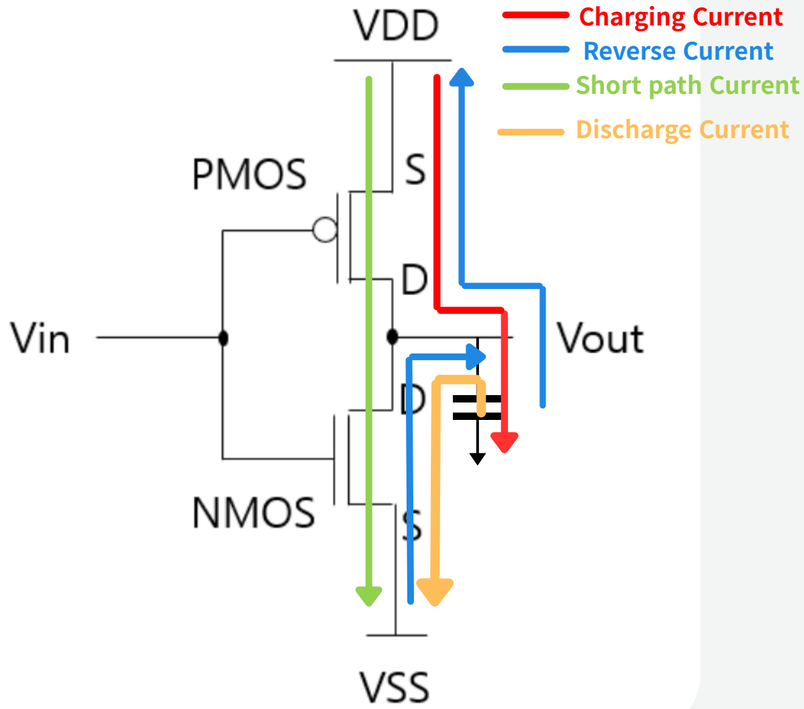

**Charge current is miximized, when short circuit is opened**



# Current Analysis - Current Analysis



# Conclusion

# Repeater 설계 및 PPA 분석 Conclusion

| PPA & Parameters | Technology Node | Multiplier | | Corner |
|---|---|---|---|---|
| | 180nm → 90nm → 45nm | 1x → 2x → 4x | | SS(85℃) → NN(27℃) → FF(-25℃) |
| | | Same load | Multi load | |
| Power<br>(IDD, ISS) | 감소 | 증가 | | 감소 |
| Performance<br>(Rise&Fall, Propagation Delay) | 증가 | 변화 X | 감소 | 증가 |
| Area | 감소 | 증가 | | 변화 X |
| PPA analysis | Power 감소<br>Performance 향상<br>Area 감소 | Power 증가<br>Performance 동일<br>Area 증가 | Power 증가<br>Performance 감소<br>Area 증가 | Power 감소<br>Performance 향상<br>Area 동일 |